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Research paper

Forensic genetic genealogy: A profile of cases solved

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ABSTRACT

This research undertakes a systematic review of cases cleared using forensic genetic genealogy (FGG) and has produced a dataset and annotated bibliography that can be used for further research. Data was collected to better understand the impact of FGG on a number of metrics relating to substantive, procedural, and distributive justice. FGG has been used primarily to clear cases involving serial and sexual violence against female and vulnerable victims, and in cases involving stranger victimization – cases that have traditionally been more difficult to clear. About 80% of victims were targeted for sexual violence, and about 28% belonged to social groups that are particularly vulnerable to criminal and sexual exploitation. About 79% of suspects and 48% of victims were of European ancestry, although the ancestry of many victims was unknown as they are victims of sexual assault. In the U.S., FGG investigations were primarily conducted in lower-income U.S. counties and were overrepresented in rural areas while being underused in the 50 largest metropolitan areas. The ten largest police departments in the U.S. cleared only 2% of cases and identified 5 suspects. The average time for FGG to clear a case was found to be 12.1 months. The rights of defendants to a fair trial are best secured when cases are cleared quickly and brought to trial in a timely fashion. Many of these cases had been open for many years: this allowed offenders to victimize others, while also leading to higher incidences of wrongful convictions, prosecutions, and innocent suspects being investigated by police. Rates of prior wrongful conviction in the dataset ranged from 2.8% of all crimes, to 3.7% of all homicides. Rates of wrongful prosecutions ranged from 10.1% of all crimes, to 16.5% of all rape-murders. Two persons have been exonerated as a result of an FGG investigation. FGG is advancing the aims of justice in several important ways and should be made more widely available, especially for victims who have been less well-served by the criminal justice system. Further steps to regulate FGG will be welcome in order to further these benefits while reassuring the public of the safety and security of their information.

1. Introduction

Forensic genetic genealogy (FGG), also known as Investigative Genetic Genealogy (IGG), is one of the newer techniques in the investigative tool kit, and it has proven to be useful in identifying an unknown individual who is the source of a biological sample, either one deposited at a crime scene or one that has been developed from an unidentified decedent. FGG came to the public's attention quite suddenly on April 24, 2018, when Joseph James DeAngelo – known as the 'Golden State Killer' – was arrested for a series of rapes and murders in California to which he pled guilty in June of 2020 [1].

Cases solved using FGG multiplied considerably after the arrest of DeAngelo, as have concerns and criticisms concerning this new technique – particularly regarding law enforcement use of 'Direct to Consumer' (D2C) DNA databases designed for the public to upload their DNA and search for relatives [2]. Genealogy has been referred to as the second most popular hobby in the US, and the second most popular topic for web surfing [3]. One recent survey by Consumer Reports found that 20% of respondents in the U.S. had taken a DNA test with one of the D2C DNA testing companies [4]. It is the popularity of genealogy in general, and genetic genealogy in particular, that has made FGG possible.

While some attempts have been made to compile lists of cases cleared using FGG [5,6], there has not yet been a systematic attempt to define FGG and to collect information about its use. This research was conducted to fill this void, and to shed light on the use of FGG to date. The data set is available to download on Mendeley Data, along with a User Guide that includes a detailed description of the coding of the dataset, as well as an annotated bibliography of the sources consulted and a brief digest of each case [7]. This data has been made available to assist with further research into FGG, and to provide accurate information to those seeking to better understand FGG and to formulate ethical policies governing its use. At present, there are plans to continue to collect this data and to publish annual updates.

In order to assess the impact of FGG on the broader criminal justice

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system, this research looks at three key aspects of justice: substantive justice, which means that there is effective crime control, that perpetrators of crimes are identified and brought to justice, and that evidence is brought before the courts that is sufficiently reliable, accurate, and probative so as to assist in their decision-making [8]; procedural justice, which means that procedures are transparent and fair, that innocent persons are not wrongfully arrested, prosecuted or convicted, and that police investigations respect the dignity and privacy interests of the public [8]; and distributive justice, which means that the benefits and the burdens of investigative activities are distributed fairly among members of the community [8].

This paper will first describe FGG and how it has been used to date. Next, it will describe the methodology used to gather FGG cases for inclusion in the dataset and to code the data along different metrics for substantive, procedural, and distributive justice. FGG has been used primarily to clear cases involving recidivist and serial offenders with multiple victims, and in offences involving sexual violence. Most of the cases involved charges of homicide and sexual assault, or both, and a significant proportion of cases involved violence from a stranger. This differs significantly from the picture of violent crimes that police were investigating within a similar time frame. Most of the victims and offenders in cleared FGG cases were of European ancestry. A significant proportion of victims belonged to especially vulnerable social groups, whereas few suspects did, and victims tended to live in lower-income counties in the U.S. It was also found that there were several wrongful convictions in these cases prior to their being cleared, as well as arrests and trials of innocent persons. Two individuals who were wrongfully convicted were exonerated as a result of the FGG investigation. Most cases solved through FGG remain pending before the courts. Of those cases, about 33% have resulted in adverse admissions by the accused. This means that an accused has admitted the facts as disclosed by the DNA evidence, even though they are adverse to the accused's interests. About 23% have resulted in guilty pleas, and 8% have resulted in convictions at trial. Solved FGG cases were not distributed evenly, being over-represented in rural areas, and significantly underused in the largest U.S. metropolitan areas and by the largest police departments.

FGG has therefore been used primarily to clear cases involving serial and sexual violence against female and vulnerable victims, and in cases involving stranger victimization – cases that have traditionally been more difficult than other cases to clear, both for homicides [9] as well as sexual assaults [10]. Many of these cases remained open for long periods of time: this allowed offenders who were wrongfully at large to victimize others, while also leading to wrongful prosecutions and convictions. FGG is therefore advancing substantive, procedural, and distributive justice on several fronts, and we can further improve upon these outcomes, such as by ensuring that a greater variety of victims and communities are being served by FGG.

2. What is FGG?

Forensic genetic genealogy (FGG) is an interdisciplinary practice that combines genomics and computer database technologies, as well as traditional and genetic methods of genealogical research to identify unknown individuals by reconstructing their ancestral lineages and drawing out their family trees. This technique can be used to identify living individuals, unknown decedents, or suspects who are the source of a biological tissue sample found at a crime scene. The FGG investigation will thus produce one or more candidates for identification, who are then confirmed or excluded as being the person of interest through forensic DNA profiling.

2.1. Genomics

Forensic DNA work examines hyper-variable regions of the genome known as Short Tandem Repeats (STRs) that distinguish one individual from another with a high degree of accuracy and precision. These DNA profiles can then be stored in law enforcement databases, such as CODIS, where they can be used to connect a single suspect to multiple offences, and to search against future crime scene samples [11].

An FGG investigation, on the other hand, begins by producing a DNA profile of the unknown individual using Single Nucleotide Polymorphisms (SNPs). In cases where whole genome sequencing is performed, this data is translated into a usable SNP profile. SNP profiles identify variation at different areas of the genome, examining about 700,000 to 800,000 SNPs [6,12].

2.2. Computer database technologies

The SNP profile of the person of interest is uploaded into one or more DNA databases that use a set of algorithms designed to match relatives. The D2C databases that are being used for FGG use proprietary algorithms to match segments, which is a limitation of this technique when it is being used as evidence in criminal cases [13]. The algorithms examine the number and length of matching DNA segments which are then used to assess the degree of relatedness between the person of interest and those persons who have uploaded their profile into the database [13,14].

At present, about 60% of searches concerning a person of interest of European ancestry will result in a match to at least a third cousin; often, this will be a sufficiently close relationship to identify the person of interest [15]. This is the rate-limiting step of FGG: a successful identification through FGG depends mainly on the ability to match the suspect with the DNA of a close relative [15]. A database that covers about 2% of the population has a very high probability of disclosing a 3rd cousin relationship to any person – provided that the samples for the database have been chosen at random [15]. A database of this coverage is foreseeable in the near future, given the continued popularity of genetic genealogy [15].

Most cases included in this dataset used the GEDmatch or FamilyTreeDNA (FTDNA) databases, where users are able to opt-in and out of law enforcement matching for criminal offences in their account settings. In North America, FTDNA will permit law enforcement searching unless users specifically opt-out in their account settings [16], while GEDmatch opts all users out of law enforcement searching in criminal investigations [17]. GEDmatch has recently modified their terms of service and privacy policy to automatically opt users in for human remains investigations whose object is the identification of a decedent [17]. This change was due in part to the difficulties faced by the parents of Wendy Stephens, a victim of serial killer Gary Leon Ridgway, known as the 'Green River Killer'. They had attempted to upload a DNA profile to GEDmatch to locate their daughter but were not opted-in to law enforcement matching [18]. These policies have been criticized by scholars as not ensuring that users have the knowledge they need to give informed consent on this issue and to opt-in and -out of law enforcement matching as they may choose [2,13].

2.3. Genealogy

Once a relative match is identified using a DNA database, genealogists then use traditional sources of information to draw out family trees to identify the person of interest [14]. These may include birth, death and marriage certificates, obituaries and grave records, and other sources of publicly available information. Once one or more relative matches have been made in the database, tracing out the family trees is a time-consuming and non-trivial task [15].

Investigations are best able to advance when a match is found with a relative who is a 3rd cousin or closer. DNA-matching algorithms often look for segments of DNA that two individuals share in common. These individuals likely inherited these shared segments through their descent from a common ancestor, and such segments are known as 'Identical by Descent' (IBD) [6,19]. Genetic genealogists then identify clusters of shared matches and work with these genetic networks to identify the most recent common ancestor (MRCA) of each cluster. The MRCA is the

most recent ancestral couple from whom the individuals are descended [6]. Once a MRCA has been found, genealogists build out the family tree to identify all the descendants of the MRCA. The person of interest will be normally found among this group of persons [6].

There may be cases where a sufficiently close match cannot be found, or where the family tree may be difficult to construct due to non-parental events, adoptions, or endogamy. In such cases, investigators may request DNA samples and other information from relatives of the person of interest [6]. Such 'targeted kinship testing' should be conducted with due attention to the ethical and privacy issues involved in the police collection and storage of the DNA samples of those who are not suspects, as recommended by SWGDAM [14]. This includes informed consent for potential relatives who are approached for a DNA sample [20], as well as the requirements that these samples later be destroyed and not uploaded into a law enforcement database [14].

Further investigative methods are used to narrow down this list. Forensic phenotyping, of both visual characteristics and biogeographical ancestry, can be used to confirm or exclude certain hypotheses, and to narrow the candidate pool [15]. In some cases, the use of information from the Y-chromosome or mitochondrial DNA can provide information as to the male and female lineages respectively, and even to identify a possible surname for the person of interest, as was done by Colleen Fitzpatrick in the investigation of the 'Phoenix Canal Killer' [21]. An FGG investigation may be able to narrow down the candidate pool to a single sibling cohort. When combined with other techniques, such as geographical profiling and non-forensic evidence, the suspect lists can be narrowed down to a very small group composed of one or a few individuals [15].

2.4. Forensic DNA profiling

Once the FGG investigation has produced a list of candidates for identification, this information is confirmed using a variety of methods. For criminal investigations looking to identify the source of a crime scene sample, this will involve matching the suspect's forensic STR-DNA profile to the crime scene DNA to exclude or include the suspect [14]. In the case of decedents, confirmation is generally made through kinship testing with a close living relative. Other methods of confirmation used in this dataset include comparing the STR-DNA profile obtained from the crime scene to a sample taken at an autopsy or from the exhumed remains of a deceased suspect.

3. Methods

This research attempted to collect all publicly available information concerning cases that were successfully cleared through FGG, going back as far as possible in time and ending on December 31, 2020. Cases were selected for inclusion in the sample through internet searches, discussion boards related to genetic genealogy, missing persons, and cold cases, as well as press releases and blog posts from private companies and non-profit organizations known to perform FGG, publications devoted to advances in forensic science, and published court records. There are about 880 news articles, blog posts, press releases and court records described in the annotated bibliography that were used to gather the data contained within the dataset. Newspaper articles are a useful resource for criminologists to track criminal activity and changes in policing over time [10]. Newspaper reports may contain information that is available nowhere else [10], particularly when a case is still pending before the court or has not come to court. Attempts were made to confirm each piece of data collected through three separate sources, whenever this was possible.

This research was not able to uncover all cases cleared using FGG. The cases collected in this dataset are likely biased in favour of cases that were well-publicized, and that were conducted by the larger forensic organizations, including Parabon Nanolabs (172 cases), Barbara Rae-Venter (87 cases), the DNA Doe Project (32 cases), Advance DNA (10 cases), Othram Inc. (9 cases), Bode Technologies (5 cases), and United Data Connect (5 cases). This data set only included cases that had been successfully cleared, and cases for which a public agency had specifically confirmed that FGG had been used in the investigation. This data therefore excludes cases in which law enforcement agencies have not come forward with information about the investigative methods they have used to clear a case. The data may be biased in favour of cases involving serious crimes, such as homicides, that are better publicized, and more likely to have excluded cases in which the investigation is not publicized for privacy reasons – such as the identification of unknown decedents, living does, or cases where an investigation is ongoing. This data also excludes FGG investigations that have yet to be cleared, as this information is not publicly available.

A 'case' for the purposes of this dataset is defined as the identification of a decedent or a living doe and, in the case of criminal investigations, a victim-offender pair. This enables data collection on individual victims, as well as the prevalence of serial/recidivist murderers and sexual offenders. Each case was assigned a unique Case Identification Number (Case ID). Cases were included from all countries which have used FGG to date, although most data analyses focus on U.S. cases only. The information on FGG cases was entered into the database by a single researcher, and so there were no issues concerning inter-coder reliability.

Cases were chosen for inclusion in the sample only if a public authority had confirmed that FGG was used to clear the investigation, and the investigation met the following definition of FGG:

1. **Investigative body:** The investigation was led by a public, and not a private, authority, such as a law enforcement agency or a coroner/medical examiner's office. Cases that are initiated and led by private individuals and organizations, even though they may uncover criminal activities – such as child abductions or incest – are not included in the dataset.
2. **Investigative techniques:** The investigation must use the techniques of genomics, computer database technologies, and traditional genealogy, as outlined above. Cases were included when family lineages were drawn out to at least three or more generations. The sample includes investigations that use Y-chromosome STR profiles and/or mitochondrial DNA profiles to trace the paternal and maternal lineages of persons of interest. Cases solved through familial searching of law enforcement databases were excluded from the sample, as this technique raises separate legal and ethical issues [22].
3. **Investigative purpose:** The investigation must be for a forensic purpose, meaning that the evidence thus produced is relevant to an ongoing legal dispute. Most cases considered here are for the identification of perpetrators of criminal offences or the identification of decedents but can also include the identification of living does. Not included in this dataset are cases that are of purely historical interest, or the identification of unknown soldiers by military authorities. Two cases have been included that were investigated as ongoing criminal matters but were found to be historical cases upon a successful resolution. These cases were not included in data analyses. The term 'forensic genetic genealogy' has been chosen for this project in preference to the term 'investigative genetic genealogy' (IGG), which has also been used [14], to highlight the fact that these investigations are directed towards an ongoing legal dispute before the public legal system.
4. **Causation:** This database only includes cases that have been successfully cleared using FGG. The resolution of any investigation – particularly very large and lengthy investigations as were many of the cases described in this dataset – has many causes. To be included here, the FGG investigation must have played a significant causal role in the resolution of the case.

4. Data collection

4.1. Substantive justice

To assess the impact of FGG on substantive justice, this research looked at the types of crimes that have been cleared using FGG, and the rates of recidivism of the suspects thus identified. Data was collected on the basic demographic characteristics of victims and offenders. These include age at time of incident, the gender, and the ancestry of the individual. Ancestry is used here to describe the geographical region(s) where the individual's recent ancestors lived. In this research, ancestry is divided up into broad, continental regions. This enables data gathered in different societies, regions, and countries around the world to be gathered and coded in the same manner for all cases. Ancestry does not necessarily correlate with an individual's self-defined ethnicity.

Data was gathered concerning whether a suspected perpetrator was a serial/repeat offender to gauge the rates of specific recidivism of offenders in the sample. The definition of a serial killer used is that given by the FBI as the "unlawful killing of two or more victims by the same offenders(s), in separate events" [23]. There must be a cooling-off period between the killings. These cases were coded as serial murders and/or serial sexual offenders. Cases in which a suspect is accused of killing two victims in a single incident were coded as multiple murderers.

Data was also gathered on whether victims and offenders belonged to social groups that are especially vulnerable to violent crimes and sexual exploitation. This category includes children and teenagers under the age of 19 years, elderly persons over the age of 65 years, persons with physical or mental disabilities, persons engaged in sex-work, as well as trans- and gender-non-conforming persons. Data was also gathered on whether the suspect was a member of a public security force, such as the police or military.

This research also examined the identification of decedents, and how effective this identification was in clearing cases in which the decedent is thought to be the victim of a homicide.

4.2. Procedural justice

This research examined the dispositions of charges in cases that are pending before the courts, whether courts had admitted the evidence produced by the FGG investigation, and whether the accused has admitted adverse facts as disclosed by the DNA evidence. Adverse admissions usually took the form of an admission that semen or other bodily fluids located at the crime scene of a sexual assault belonged to the accused, or that the accused was the parent of an infant who was found deceased. This data helps to gauge the admissibility of FGG evidence before the courts, and how effective it is in securing convictions. This data will be updated as cases move though the trial and appellate courts.

This research examined the number of persons who had been wrongfully arrested, tried, and convicted before a case was successfully cleared through an FGG investigation. Data was also collected on exonerations achieved to date using FGG.

Data was gathered on the dates of the incident, the FGG investigation, and case clearance so that a within-group comparison could be made of the length of the investigation versus the length of the FGG investigation.

Data was gathered, where publicly available, about the tissue source of the sample used in the FGG investigation, the DNA databases consulted for relative matching, as well as the methods of confirmation used to confirm the results of the FGG investigation – such as matching with a living relative, a tissue sample taken at an autopsy or prior medical procedure, or surreptitious sampling of discarded items by police.

4.3. Distributive justice

The burdens and benefits of any criminal justice practice or policy should be distributed fairly among members of society. There are numerous parameters that are relevant for assessing distributive justice, and this paper will only cover a small number of these.

Data was collected on the distribution of FGG cases in the U.S. using data on Metropolitan [24] and Micropolitan [25] Statistical Areas from the U.S. Census Bureau. This data assesses how FGG cases are distributed among communities of different population sizes, and among the largest metropolitan police departments. This data was then compared with rates of violent crimes, including homicides and sexual assaults, as provided in the FBI's Uniform Crime Reports for the years of 2018 and 2019 [26–29]. This enables a comparison between cases cleared using FGG with the pool of violent criminal offences that police departments were investigating in a similar time period, and to determine how the FGG cases differed from other violent crimes.

Demographic information on ancestry was collected for both victims and offenders. In the case of offenders, this information was readily available. The ancestry of victims could not be determined in the majority of cases, as they are victims of sexual assault whose privacy is being protected. As a result, this data is not able to reliably assess whether there are racial and ethnic disparities among the victims who are being served by FGG investigations, although data from known cases suggests that there may be such disparities.

Information was also collected on the per capita income for all U.S. counties that had conducted an FGG investigation [30], to assess the socio-economic status of the victims and communities that are being served by FGG, and whether higher-income counties are making more or less use of FGG. (Figs. 1–6).

5. Result

5.1. Overview of FGG Investigations

By far most cases of FGG were conducted in the U.S. Police in Canada have publicly acknowledged solving three cases to date (Case IDs 1308, 1396, and 1416). There was one case involving a double homicide in Sweden that took place in 2004 (Case IDs 1375–1376) [31,32]. This case was used as a pilot study to assess the feasibility, as well as the ethics, of

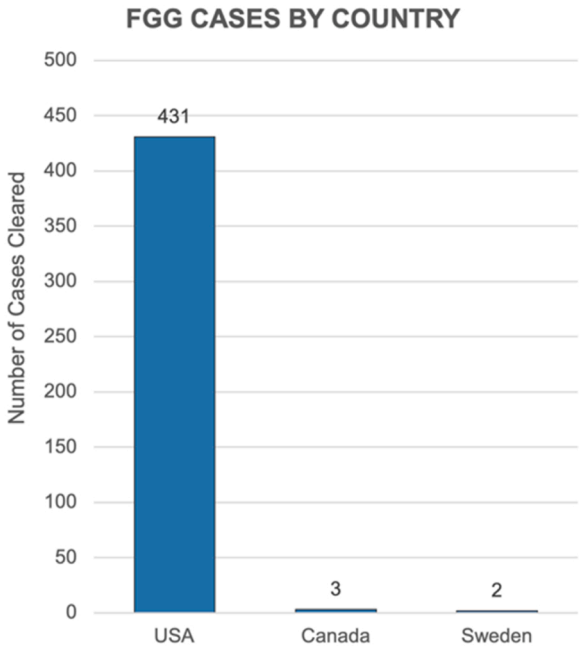


Fig. 1. FGG cases by country. This figure includes data from all cases from all countries, out of a total of 436 individual cases. A case is defined as the identification of a living doe or a decedent, and for criminal investigations it is a victim-offender pair.

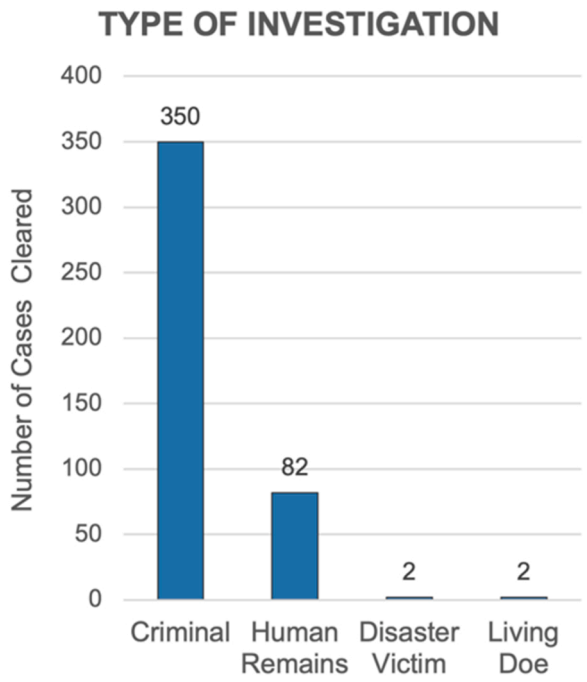


Fig. 2. FGG cases by type of investigation. This figure includes data from all cases from all countries, out of a total of 436 individual cases. A case is defined as the identification of a living doe or a decedent, and for criminal investigations it is a victim-offender pair.

using FGG to solve Swedish cases [31]. As a result, further cases are being assessed and Sweden plans to draft national guidelines covering regulating the practice of FGG [31].

To date, FGG investigations fall into four broad categories: 350 cases (80%) are criminal investigations, and 82 (19%) are investigations to identify unknown decedents. There have also been 2 cases that successfully identified living does (Case IDs 1221 and 1228), as well as 2 victims of a mass disaster (Case IDs 1145 and 1146), both of whom were killed in the Paradise Camp Fire in California in November of 2018.

The first FGG cases in the U.S. were cleared in January of 2015 in Arizona when Colleen Fitzpatrick identified Bryan Miller as the lead suspect in the 'Phoenix Canal Killings' using Y-STR technology (Case IDs 1003–1004). This is the only case in the dataset to use only Y-STR matching to successfully clear a case. The first FGG investigation using genealogical matching of autosomal DNA in GEDmatch was solved by Colleen Fitzpatrick and Margaret Press of the DNA Doe Project when they successfully identified Marcia King as the decedent known as the 'Buckskin Girl' (Case ID 1016) [33]. FGG investigations have been successfully conducted in 43 out of 50 U.S. states, as well as the District of Columbia. Most cases were cleared in California, many of which involved the successful identification and prosecution of Joseph James DeAngelo, known as the 'Golden State Killer', for a series of rapes and murders for which he has pled guilty (Case IDs 1017–1088). California also successfully identified Roy Waller as the suspect in a series of sexual assaults against nine women (the 'Nor-Cal Rapist'), for which he was convicted at trial (Case IDs 1112–1120). As of December 31, 2020, California had cleared a total of 129 cases, having identified 29 individual suspects of criminal offences, as well as 5 decedents, 2 victims of mass disasters, and 2 living does.

By far most cases involved a homicide and/or a sexual assault. The less serious charges (attempted murder, robbery, kidnapping, desecration, aggravated assault, assault, weapons, child abuse, and obstruction) nearly always took place in conjunction with a homicide or a sexual assault.

5.2. Dispositions of FGG investigations

Most cases that have been cleared by FGG are pending before the courts (39%), and the second largest category are cases that have been cleared by exception due to the death of the offender (19%). A total of 95 cases (13%) have resulted in a guilty plea, and 34 cases (5%) have received convictions at trial. A total of 55 cases (8%) have been barred by statutes of limitations, all of which involved charges for sexual assault.

Of the live charges that are currently before a court, 69% are pending. Of the cases that had completed by the end of the data collection period, 23% resulted in a guilty plea and 8% resulted in a conviction at trial. No cases to date have received an acquittal. One case has been dismissed: Erlene Gayle Lee (Case ID 1261) had been charged with a co-accused, Willie Moore (Case ID 1262), for the murder of Anthony Baltes in Tulsa, Oklahoma in November of 1983; the charges against her were dismissed, and she was expected to testify on behalf of the state [34]. However, Moore passed away in custody in December of 2020, and the case has now been closed [35].

Fig. 7 indicates whether an accused in a case pending before a court in the U.S. has made adverse statements admitting the facts disclosed by the DNA evidence, either in a statement or by entering a guilty plea. This data looks at individual suspects rather than at victim-offender pairs in order to remove duplicate data from suspects with multiple charges. About 1/3 of suspects (33%) have made adverse admissions regarding the DNA evidence in the case. Of the 33 accused who have made adverse admissions, 10 cases remain pending (30%), and 23 (70%) have resulted in convictions – 20 cases through guilty pleas (61%), and 3 accused who were convicted at trial (9%).

Fig. 8 examines all cases that are pending before a U.S. court. About 27% of U.S. courts have admitted the DNA evidence of identification of the suspect as obtained through the FGG investigation; this includes cases which concluded through a guilty plea, as well as those defendants who have been convicted at trial. Most cases (71%) are pending, and no decision has yet been made by the court regarding the admissibility of the evidence obtained through FGG. No court had ruled to exclude evidence obtained through FGG during the data collection period.

5.3. The types of offences being solved by FGG

Fig. 9 examines individual victims of criminal offences in the U.S. About 69% of victims were victimized by a serial or suspected serial offender. About 37% of victims had been victimized by a suspect who has been convicted of, or who is charged with, at least two or more sexual offences committed in separate incidents. A total of 31% of victims had been victimized by a suspect who has been convicted of, or who is charged with, more than one homicide committed in separate incidents; most were victimized by a serial murderer who was motivated by sexual violence. About 1% of victims were victimized by a multiple murderer, who has been convicted of, or is currently charged with, the murder of two persons in a single incident. For 31% of victims, there was no data that would indicate that the suspected perpetrator is a serial or multiple offender; these offences are thought to have been committed by a non-recidivist offender.

There were 15 murderers of newborns or infants whose age could not be determined. Of the 13 accused charged with the murder of an infant or newborn, 4 (30.8%) were serial offenders who are thought to have killed more than one child, and one was a multiple murder who killed two children in a single incident (7.7%). The remaining 8 (61.5%) are not known to have killed more than once.

As can be seen from Fig. 10, a total of 75% of all U.S. suspects in the dataset were motivated by sexual violence, and 80% of all U.S. victims of crime were victimized by an offender whose motivation was sexual violence. This indicates that law enforcement personnel who are making decisions to use FGG as an investigative tool tend to choose cases with multiple victims who have been subjected to sexual violence.

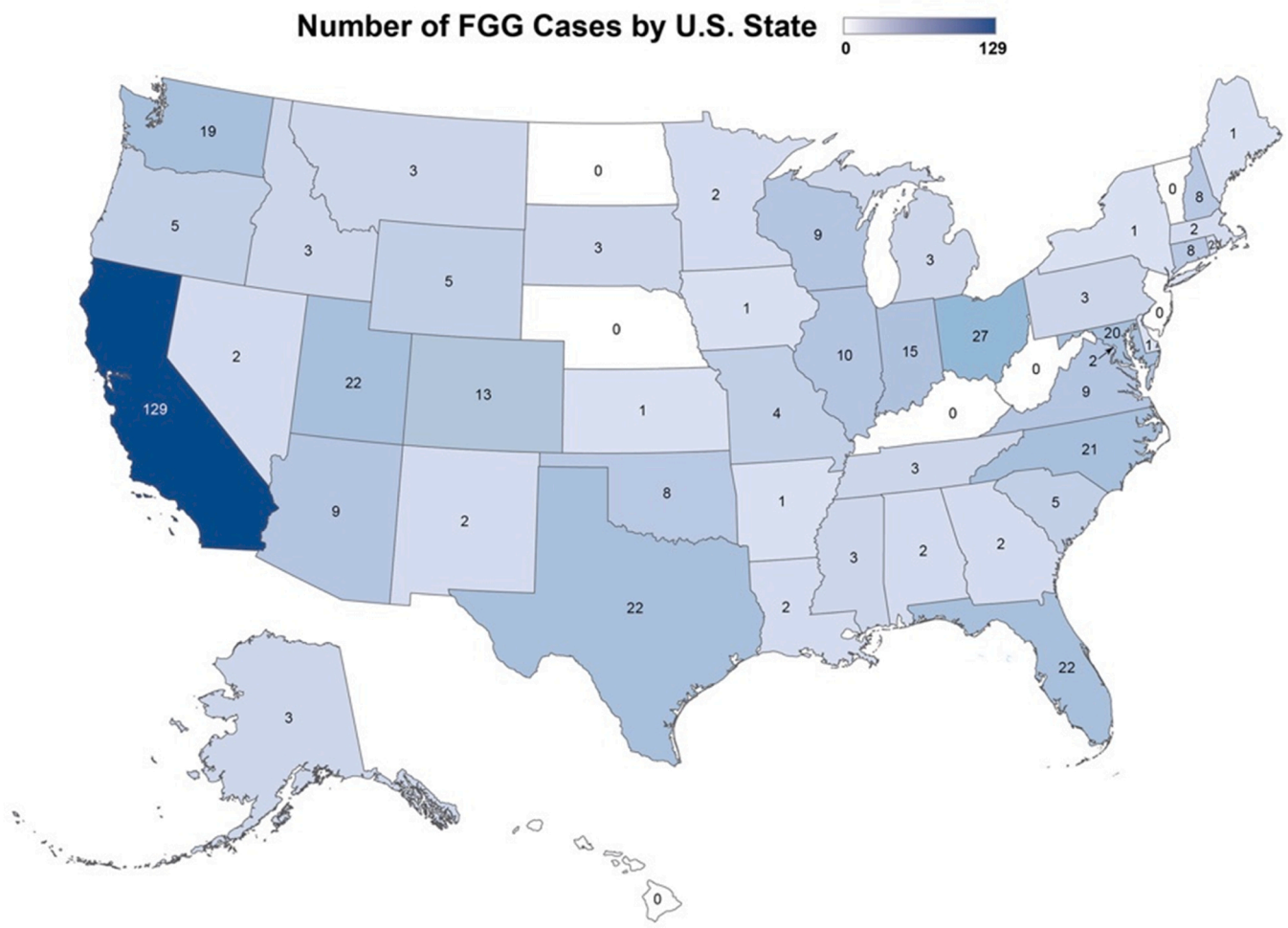


Fig. 3. Number of FGG cases by U.S. state. This map includes all FGG cases solved in the U.S. up to December 31, 2020, based on the state in which the case was cleared.

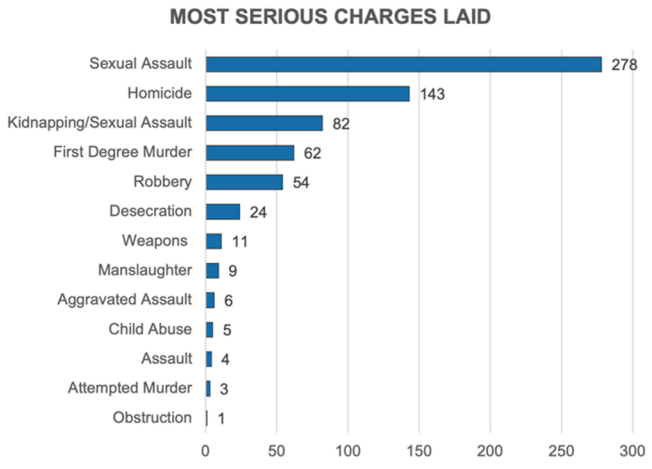


Fig. 4. Most serious charges laid. Data was collected for the two most serious charges in all cases, out of a total of 682 charges (in some cases, there was only one charge). If a sexual assault occurred, this was included as the most serious charge; if a sexual assault took place in conjunction with a homicide, then it was always included as the second most serious charge. Data for U.S. cases only was examined and found to be not significantly different from the overall data presented here, due to the small number of non-U.S. cases. All cases of manslaughter in the dataset are non-negligent manslaughter.

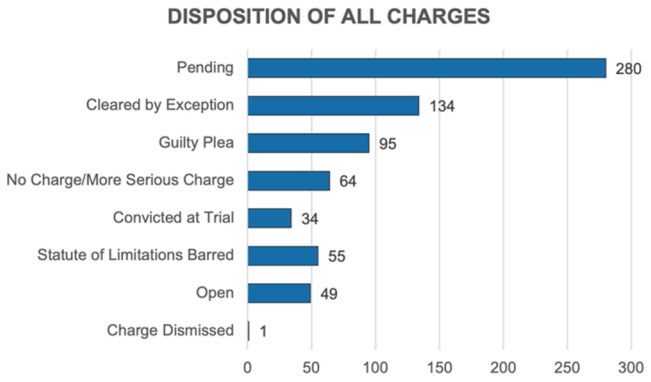


Fig. 5. Disposition of all charges. This figure includes all cases involving a criminal offence, from all countries, including open homicide cases in which a decedent has been identified but not a perpetrator, out of a total of 712 dispositions. For criminal cases, it includes cases by victim-offender pair. The data for U.S. cases was also examined and the results were not significantly different from all cases, due to the small number of non-U.S. cases.

Fig. 11 looks at whether victims and suspected perpetrators belong to a group that is especially vulnerable to criminal and sexual violence and exploitation. Victims and suspects were coded as being especially vulnerable when they were aged 19 and under, aged 65 and older, were identified as being physically or mentally disabled, were identified as

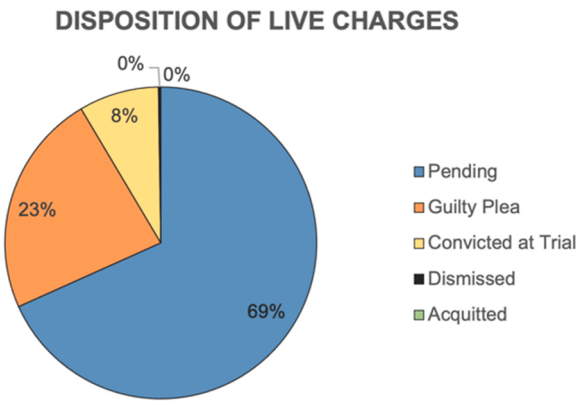


Fig. 6. Disposition of live charges. This figure includes all charges that are currently before a court, for all cases and all countries, out of a total of 410 live charges. For criminal cases, this data includes the disposition of live charges for each victim-offender pair, and so includes multiple charges for recidivist offenders. Note that charges and their dispositions are very similar when only U.S. data is included, due to the small number of non-U.S. cases.

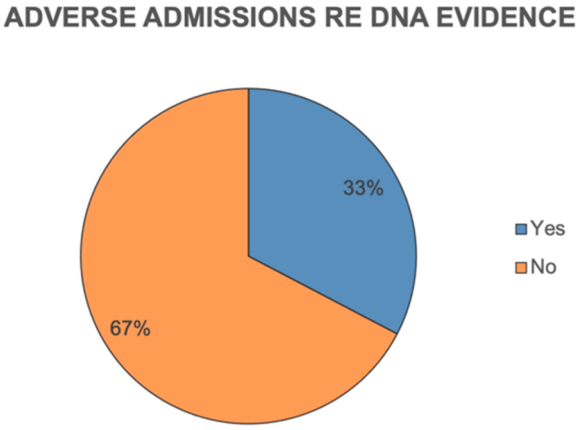


Fig. 7. Adverse admissions re dna evidence, per accused This figure includes data for U.S. cases where charges are pending before a court, per individual accused, out of a total of 101 accused. It indicates whether the accused has made an adverse admission regarding the DNA evidence gathered in the case.

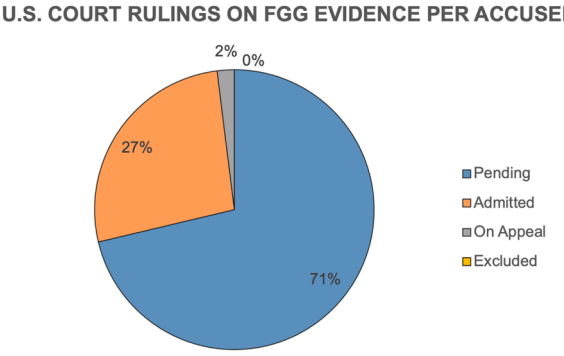


Fig. 8. U.S. court rulings on FGG evidence per accused. This figure includes data on all cases that are pending before a U.S. court, per individual accused, out of a total of 101 accused.

being sex workers at the time they were killed, or who appear to have self-identified as being trans or gender non-conforming persons. About 28% of victims belonged to an especially vulnerable group, mostly due to their age, whereas only 8% of suspects did – all of whom were so

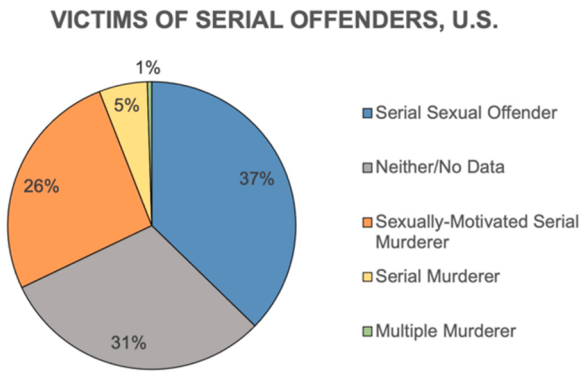


Fig. 9. Victims of serial offenders, U.S. This figure looks at U.S. cases only and examines whether the victim of a crime has been victimized by a serial offender. This data is out of a total of 405 individual victims and includes suspected serial offenders with charges currently pending before a court. The definition of a serial offender is that given by the FBI: two or more offences committed in separate incidents with a cooling-off period in between [23]. The category labelled 'neither/no data' includes cases in which the suspect has no previous convictions for homicide or sexual assault, as well as cases on which there is no data on the suspect's prior criminal history.

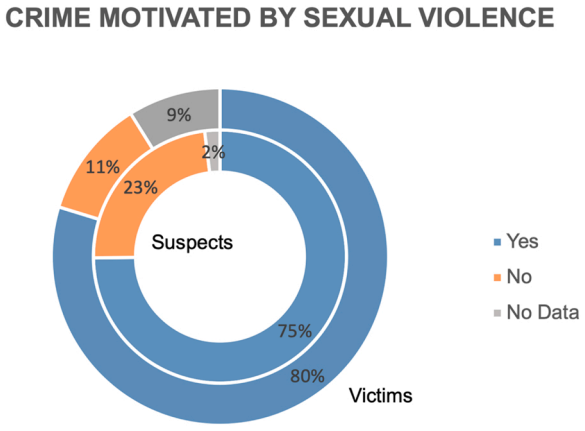


Fig. 10. Crime motivated by sexual violence. This figure looks at U.S. cases only and compares criminal offences that were motivated by sexual violence with those that were not. This is broader than whether or not charges were laid for a sex offence: there are many cases in which charges of sexual assault are not laid, perhaps because the statute of limitations has run out, the victim was already deceased when the sexual offence took place, or the sexual offence was interrupted. This data is out of a total of 405 individual victims, and 155 individual offenders.

coded because they were under the age of 20 years at the time of the criminal offence.

It was also found that about 11% of suspected perpetrators had been members of a public security force – either the military, police, or a fire department – whereas less than 1% of victims were. Because many of these suspects were serial offenders, they constituted 29% of all criminal investigations in the sample. In some cases, this status may have played a role in the criminal offending, either by allowing the offenders to use this status and the skills they learned to overpower their victims and avoid detection, or by asserting their credibility and deflecting suspicion.

Joseph DeAngelo (Case IDs 1017–1088), for example, was at one time a police officer, but there were several other suspects in the dataset who also leveraged their status as members of security forces to facilitate their criminal offending. Angelo Alleano Jr. (Case IDs 1369–1372) has been charged with four home invasion rapes in Manchester, Connecticut that took place between 2001 and 2008. A profile of the offender was

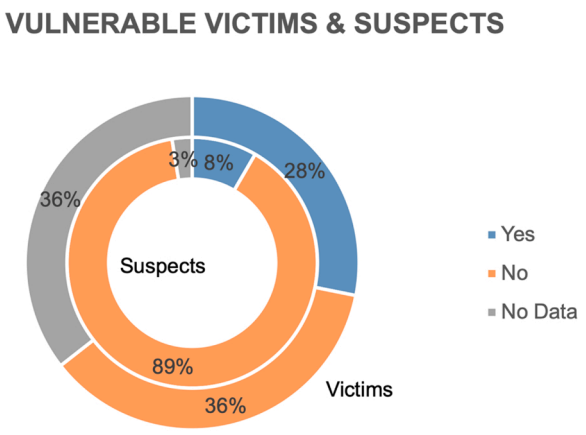


Fig. 11. Vulnerable victims & suspects. This figure includes all U.S. cases involving criminal activity, out of a total of 155 individual suspects and 405 individual victims. It examines whether victims and suspects belong to groups that are especially vulnerable to criminal offences and sexual violence. Victims and suspects were coded as being especially vulnerable when they were aged 19 and under, aged 65 and older, were identified as being physically or mentally disabled, were engaged in sex work at the time they were killed, or were identified or appear to have self-identified as being trans or gender non-conforming persons.

obtained from the FBI's Violent Crime Behavioral Analysis Unit in 2008 which indicated that the offender might be a police officer or a firefighter [36]. The profile recommended taking DNA samples from local police and fire departments, but this suggestion appears to have been dismissed by investigators at the time [36].

William Baer and his then-spouse Melissa Jo Schafer (Case IDs 1391–1392) are charged in the May 17, 1999, home invasion robbery and murder of 39-year-old Saad Kawaf in Jacksonville, Florida. Baer worked for the Jacksonville Sherriff's Office as a detective for 27 years before recently retiring [37,38]. At the time of the murder, Baer was engaged in surveillance against Kawaf as part of an ongoing drug investigation [37,38]. Baer thus learned that Kawaf had yet to make a weekly bank deposit of $30,000 and was keeping that cash in his home [37,38]. Many of the cases that Baer worked on in his 27-year career will now have to be reviewed for official misconduct [37].

Ivan Wayne Linebaugh was arrested on March 30, 2020, and charged in the April 25, 2000, sexual assault of a female in Madisonville, Texas (Case ID 1351). The woman reported being sexually assaulted by a police officer at a traffic stop, who threatened her with jail if she did not do as he told her [39]. At the time of the assault, Linebaugh was a Madison County Sheriff's Office Sgt. Detective, and was a candidate for Madison County Sheriff in 2016 [39].

Table 1
Serial homicides and sexual serial homicides in the U.S.

Type of Homicide	Number of Homicides, 1960–1989	Probability	Expected number	Observed number
Serial Homicide	2028	0.043336326	7	11
Sexual Serial Homicide	1602	0.032279398	5	49
Other	46302	0.924384276	148	100
Chi-Squared Value, df= 2	**405.0533**			
p-Value at 99%	< 0.00001			
Significance	SIG			

5.4. Proportion of serial offences compared with all serial offences in the U.S

Table 1 examines data about the proportion of serial homicides and sexually motivated serial homicides (rape-murders) among the U.S. cases in the dataset and compares this with data about the proportion of such crimes in the U.S. over a similar time period. A serial homicide is at least two murders committed by the same perpetrator in separate incidents [23]. A serial sexual homicide is an offence committed by a serial homicide offender who has raped or sexually violated the victims, either before or after death, or in which sexual violence was a prime motivating factor for the offender. This latter category captures cases in which the sexual offence could not be completed, was only completed after the victim's death, or in which the offender killed one or more third parties, such as friends or family members, before sexually assaulting the victim. This data is compared with Missen's Table 6A [40] in which he provides data about serial homicides and sexual serial homicides in the U.S. from 1960 to 1989. Missen does not include data from 1990 to the present, whereas the cases in this dataset go from 1963 to the present day. The number of serial killers operating in the U.S. increased to a high in the 1970 s and 1980 s before declining again [41]. As serial homicides have generally decreased since the early-mid 1990 s, the true number of expected serial homicides is likely lower than the one estimated here and likely has little effect on the significance of the findings.

Fig. 12 and Table 2 examine the relationship of victims to offenders in the data set and compares this to the expected relationships based upon the FBIs UCR data for 2018 and 2019 [26,27]. This data includes all U.S. cases that involved a homicide, by victim-offender pair, and out of a total of 160 homicides. This allows for a comparison between the cases solved through FGG and the general pool of homicides being investigated by police departments in a similar time period.

As can be seen, there were far more homicides than expected that were committed by a stranger to the victim (117 vs. 16 expected), and far fewer homicides committed by acquaintances (4 vs. 77 expected), partners (2 vs. 15 expected) and other family members (4 vs. 9 expected). There were more murders of children committed by a parent in this sample than expected (18 vs. 5 expected), due to the number of cases of neonaticide and infanticide that have been solved using FGG. This supports the hypothesis that FGG is being used to solve cases that have proven difficult to solve through traditional investigative methods, such as stranger murders and infanticides.

Fig. 13 and Table 3 examine the motivation of suspected perpetrators

Table 2
Relationship of victim to offender, expected vs. observed.

Relationship of Victim to Offender	FBI UCR 2018–2019 Average	Expected probability	Expected number	Observed number
Unknown	7107.5	0.4822	77	4
Acquaintance	1922	0.3104	21	15
Other- Known to Victim	1540.5	0.1045	17	0
Stranger	1482.5	0.1006	16	117
Partner	1386	0.094	15	2
Other Family	843.5	0.0572	9	4
Child	458	0.0311	5	18
Chi-Square, df= 6	**773.3291**			
p-Value, 99%	< 0.00001			
Significance	**SIG**			

Fig. 12 & Table 2. Relationship of victim to offender, expected vs. observed. This includes data from all U.S. homicide cases, out of a total sample of 160 U.S. homicides, and compared with the FBI's UCR data on homicides in the U.S., 2018–2019 [26,27]. This enables the FGG homicide investigations to be compared with the overall pool of homicide investigations facing U.S. police forces in about the same time period (2020 data was not yet available at the time of writing).

Table 3
Motivation of offender, expected vs. observed.

Motivation of Offender	FBI UCR 2018–2019 Average	Expected probability	Expected number	Observed number
Unknown	5711	0.4029635	64	6
Rape	11	0.0007762	< 1	114
Robbery/ Burglary	622	0.0438878	7	14
Argument	3393.5	0.2394426	38	0
Domestic Violence/ Other	4435	0.31293	50	26
Chi-Square, df= 4	**115818.986**			
p-Value, 99%	< 0.00001			
Significance	**SIG**			

Fig. 13 & Table 3. Motivation of offender, expected vs. observed. This includes data from all U.S. homicide cases, out of a total of 160 homicides, and compared with the FBI UCR data on homicides in the U.S. from 2018 to 2019 [26,27]. This enables the FGG homicide investigations to be compared with the overall pool of homicide investigations facing U.S. police forces in about the same time period (2020 data was not yet available at the time of writing). Each homicide was coded as having only one main motivation. Murders of close family members that were motivated by sexual violence were classified as motivated by domestic violence, rather than rape. Domestic violence and 'other' are categorized together here, as it is in the FBI-UCR dataset. In this sample, there were 24 cases motivated by domestic violence, and two that had a motivation coded as 'other'.

Table 4
FGG Investigations by demographic area vs. u.s. population, 2019.

Demographic Area	U.S. Population, 2019	p Expected	n Expected	n Observed
Top 10 Metro	86,046,773	0.2621	113	69
Top 11–25 Metro	51,587,053	0.1572	68	75
Top 26–50 Metro	42,882,041	0.1306	56	82
Top 51–100 Metro	39,372,349	0.1120	48	40
Top 101–200 Metro	36,363,085	0.1108	48	49
Small Metro (201–384)	26,577,214	0.0810	35	24
Micro Metro (MSA 385–954)	27,252,936	0.0830	36	20
Rural Areas (MSA 955)	18,158,072	0.0553	24	71
Chi-Sq, df= 7	**133.8887**			
p-Value at 99%	< 0.00001			
Significance	SIG			

Table 4 examines the distribution of FGG investigations across different demographic areas in the U.S., by Metropolitan Statistical Area (MSA). There are 955 MSAs, including all rural areas, which were coded as '955'. This data is compared with the distribution of the U.S. population across those same statistical areas as of 2019, according to the U.S. Census Bureau [24,25]. This data includes a total of 430 FGG investigations conducted in the U.S., with criminal cases examined by victim-offender pair. There was found to be a chi-squared value of 133.8887, which was found to be significant at 7 degrees of freedom and a 99% confidence level.

of homicides in the sample and compares this with the expected motivations for homicides based upon the FBI UCR data for 2018 and 2019 [26,27]. This data includes all U.S. cases that involved a homicide, by victim-offender pair, out of a total of 160 homicides. This allows for a comparison between the cases solved through FGG and the general pool of homicides being investigated by police departments in a similar time period.

As can be seen, there was a significant difference in homicides where rape was the motivation; in the sample, rape was overwhelmingly the largest motivation of homicide offences, whereas for all reported homicides it was very rare (114 vs. < 1 expected). There was also a slight overrepresentation of robberies as a motivation for the homicide (14 vs. 7 expected). In many of the robbery-homicide cases, there was sufficient biological material at the crime scene because the victim had fought the intruder. The sample source used in the robbery-homicides was the blood of the suspect (9 cases), touch DNA (2 cases), and the sample source was unspecified in 3 cases. All other motivations were underrepresented among FGG cases, including domestic violence (26 vs. 50 expected), and arguments (0 vs. 38 expected). The types of cases being solved through FGG are therefore those in which there is a sufficient quantity of biological tissue to obtain a DNA profile.

5.5. *Demographic characteristics of victims & offenders*

Fig. 14 compares the age of the victims and the suspects at the time of the criminal offence. Only 37 suspects were over the age of 40 at the time of the offence (9.7%), and only 9 were above the age of 50 (2.3%). There is a much greater variation in the age of victims, who tend to be in their teens and 20 s, and who also tend to be very young or very old. According to Wright, the mean age of serial rapists is between 29 and 35 years of age [10], and that is consistent with the age of suspects in this dataset.

Fig. 15 compares the gender of suspected perpetrators with the gender of victims and decedents whose cases were cleared using FGG. Suspects tended overwhelmingly to be male (89%), whereas victims tended overwhelmingly to be female (84%). Of the male victims, 27 out of 64 (42%), were victimized during an incident of sexual violence against a female partner or family member. Of the remaining male victims, 9 out of 64 (14%), were children under the age of 10. Most of the remaining male victims were victimized in the course of a robbery or home invasion.

Fig. 16 compares the ancestries of suspected perpetrators of criminal offences with victims of crimes and identified decedents. Most suspects (79%) and most victims (49%) were of European ancestry, and there was a much greater variation of ancestries among victims than among suspects. A large number of victims have an unknown ancestry, as they are victims of sexual assaults whose personal information is being kept private.

Fig. 17 includes data on the ancestry of victims of crime and identified decedents. About 49% of victims are identified as being of European ancestry, about 2% of victims were identified as being of an African American ancestry, about 1% were identified as being of an East Asian ancestry, and about 3% were identified as being of another ethnic minority ancestry, including.

Indigenous, Middle Eastern, Indian Subcontinent, African, and Central/South American ancestries. Fully 45% of all victims are of unknown ancestry, and this is because they have experienced a sexual assault and their information is being kept private.

Fig. 18 indicates the ancestry of victims of crime and identified decedents for all cases where the ancestry was reported. About 89% were of European ancestry, and about 11% were of a non-European ancestry. About 3% of victims and decedents were reported to be of African American ancestry, about 2% were reported to be of an East Asian ancestry, 6% were reported to be of another ethnic minority ancestry, including Indigenous, Middle Eastern, Indian Subcontinent, African, and Central/South American ancestries. Two cases involved women of Indigenous ancestry.

Fig. 19 indicates the ancestry of suspected perpetrators of criminal offences that were solved using FGG. About 79% of suspects identified using FGG were of European ancestry, 13% were of African American ancestry, and about 3% were of another ethnic minority ancestry. In this dataset, this included suspects who were identified as being of Indigenous, Central/South American, and Middle Eastern ancestries. This may reflect the ancestries of those users who have uploaded their DNA into the D2C databases being used by FGG. There is currently little data on the ancestries of the users of these databases, and in particular those who have opted-in to law enforcement searching.

Table 5
FBI-UCR Violent crime rates by demographic area, 2018–2019 average.

Demographic Area	Number violent crimes – All	p Violent Crimes	Number Homicides	p Homicides	Number sex assaults	p Sex Assaults	Number Robberies	p Robberies
Metro Counties Over 100,000	118040	0.5245056	1618	0.4265034	13936	0.42650344	23492	0.77669775
Metro Counties 25,000–99,000	42296	0.1879405	598	0.2015608	6586	0.20156083	3361	0.11112213
Metro Counties Under 25,000	19002	0.0844346	234	0.1036572	3387	0.10365723	1234	0.04079878
Non-Metro Counties Over 25,000	18946	0.0841857	306	0.1024943	3349	0.10249426	1180	0.03901342
Non-Metro Counties 10,000–24,999	14812	0.0658165	256	0.0762969	2493	0.07629686	691	0.022846
Non-Metro Counties Under 10,000	11954	0.0531171	253	0.0894874	2924	0.08948738	288	0.00952192
TOTAL	**225,050**	**1**	**3265**	**1**	**32,675**	**1**	**30,246**	**1**

Table 5 looks at how violent crimes, homicides (not including negligent manslaughter), sexual assaults, and robberies were distributed across six demographic population areas according to the FBI UCR data sets for 2018 and 2019 (2020 data was not available at the time of writing). These were averaged to create the probability distributions seen in Table 5. Source: [26,27]

Table 6
Expected vs. observed violent crimes in sample versus the fbi-ucr, based on demographic area, 2018–2019 average.

Demographic Area	Expected Violent Crimes – All	Observed violent crimes	Expected homicides	Observed homicides	Expected sex assaults	Observed sex assaults	Expected robberies	Observed robberies
Metro Counties Over 100,000	253	215	81	67	114	131	40	17
Metro Counties 25,000–99,000	91	156	30	58	54	81	6	17
Metro Counties Under 25,000	41	21	12	8	28	12	2	1
Non-Metro Counties Over 25,000	41	18	15	4	27	11	2	3
Non-Metro Counties 10,000–24,999	32	35	13	5	20	19	1	11
Non-Metro Counties Under 10,000	26	38	13	21	24	14	1	3
TOTAL	**484**	**483**	**164**	**163**	**267**	**268**	**52**	**52**
CHI-SQ, df= 5	**80.6144**		**47.7993**		**38.8762**		**138.3917**	
p-Value 99%	< .00001		< .00001		< .00001		< .00001	
Significance	YES		YES		YES		YES	

Table 6 examines all U.S. cases involving a criminal offence that have been cleared using FGG and compares the rates of crime and number of FGG investigations in the sample with what would be expected given the crime rates as seen in the FBI-UCR dataset concerning violent crimes that took place in 2018 and 2019 (with 2020 data not available at the time of writing). Table 6 compares the expected versus observed number of violent crimes in the present sample based upon the probability distributions calculated in Table 5 and uses a Chi-Squared Goodness of Fit Test to assess the deviation from the expected distribution. All cases of manslaughter in this dataset included non-negligent manslaughter only. Source: [26,27].

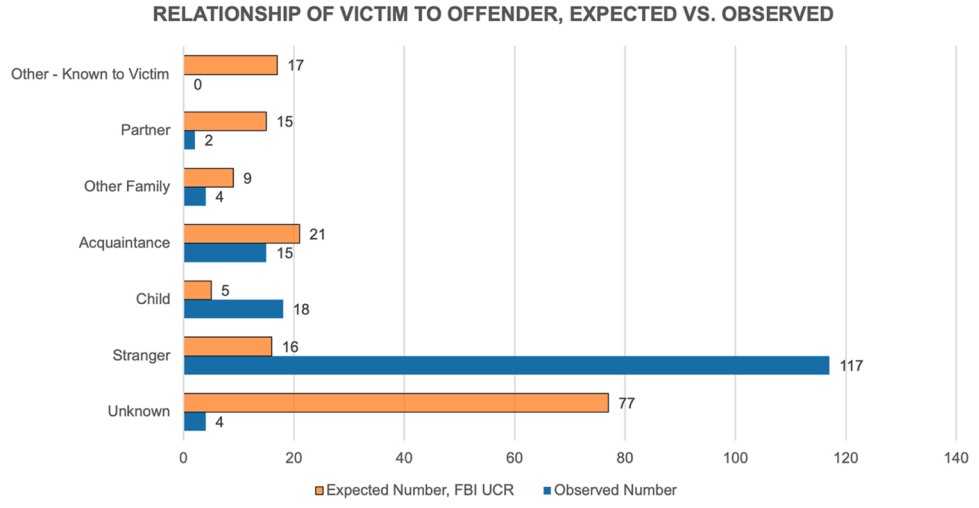


Fig. 12. Relationship of victim to offender, expected vs. observed.

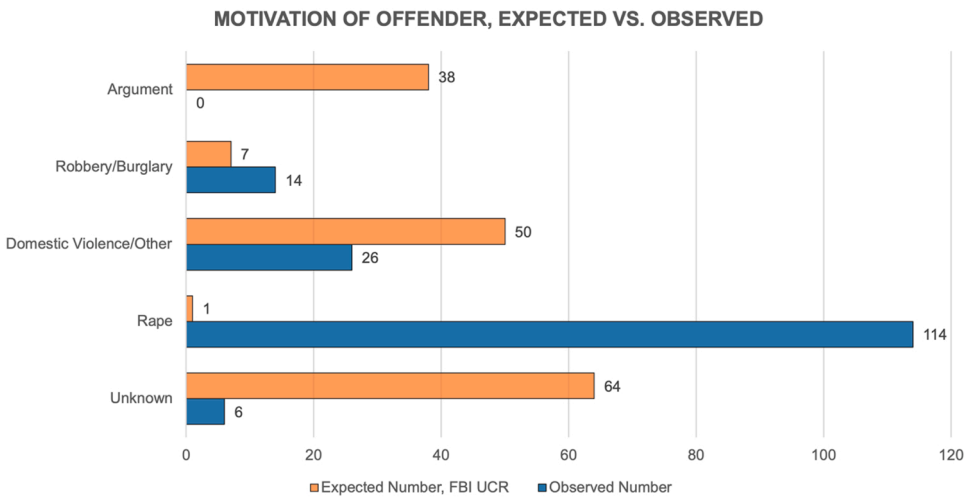
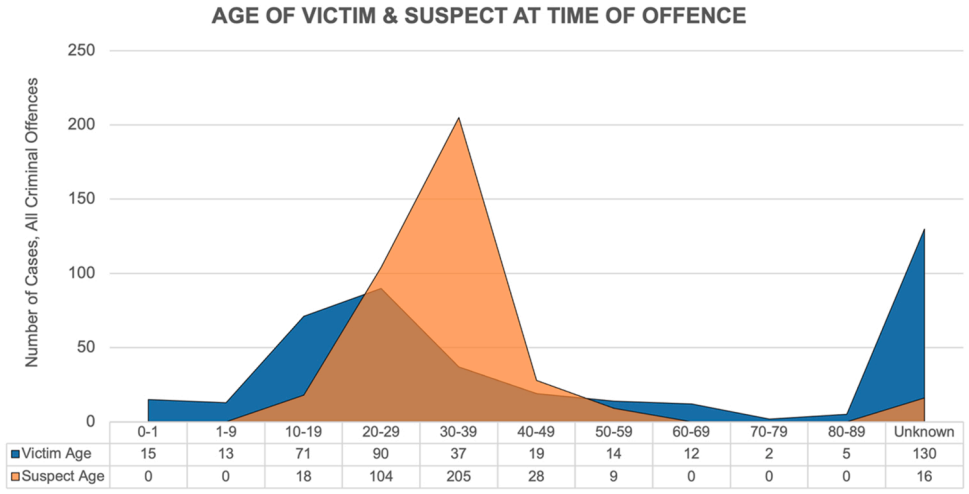
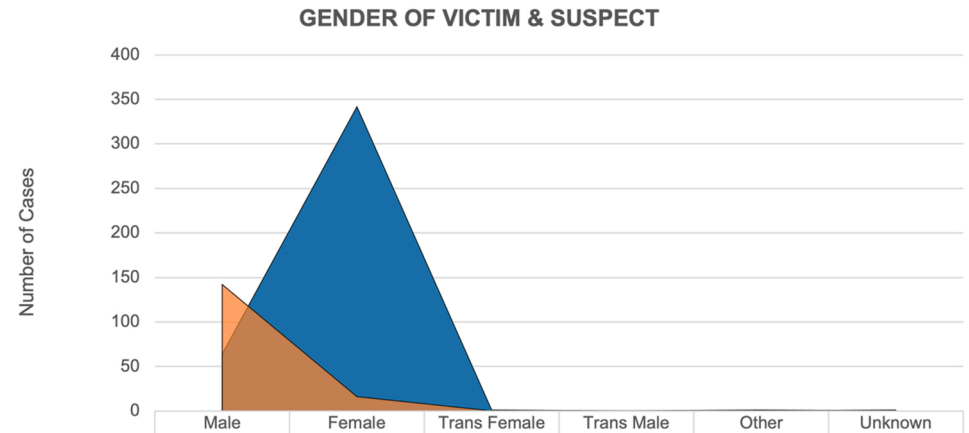


Fig. 13. Motivation of offender, expected vs. observed.



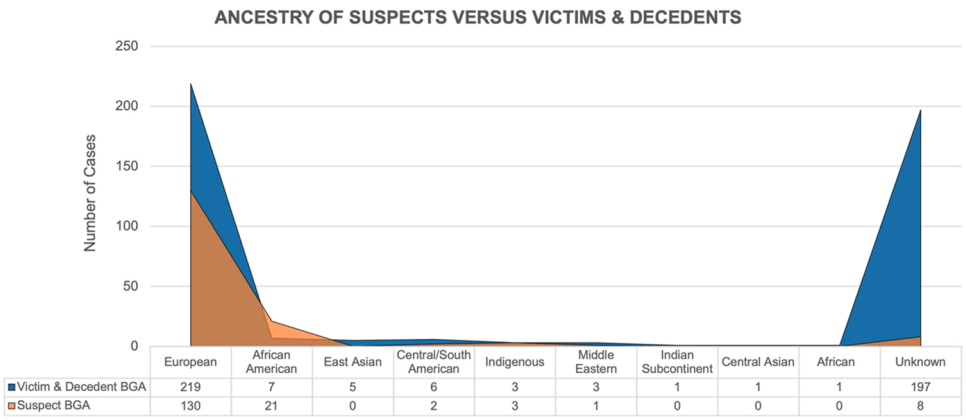
	0-1	1-9	10-19	20-29	30-39	40-49	50-59	60-69	70-79	80-89	Unknown
Victim Age	15	13	71	90	37	19	14	12	2	5	130
Suspect Age	0	0	18	104	205	28	9	0	0	0	16

Fig. 14. Age of victim & suspect at time of offence. this figure includes data from all criminal offences, for all counties, for each victim-offender pair.



	Male	Female	Trans Female	Trans Male	Other	Unknown
Victim Gender	64	342	1	0	0	1
Suspect Gender	142	16	0	0	1	0

Fig. 15. Gender of victim & suspect. this figure includes data for all criminal offences, from all countries, and for each victim-offender pair, out of a total of 408 victims and unidentified decedents, and 159 suspects.



	European	African American	East Asian	Central/South American	Indigenous	Middle Eastern	Indian Subcontinent	Central Asian	African	Unknown
Victim & Decedent BGA	219	7	5	6	3	3	1	1	1	197
Suspect BGA	130	21	0	2	3	1	0	0	0	8

Fig. 16. Ancestry of suspects vs. victims & decedents. this figure includes data for all criminal offences, from all countries, and for each victim-offender pair, out of a total of 443 victims and identified decedents, and 159 suspects. In 6 cases, suspects were identified as having two major ancestries, and these were counted separately for a total of 165 ancestries.

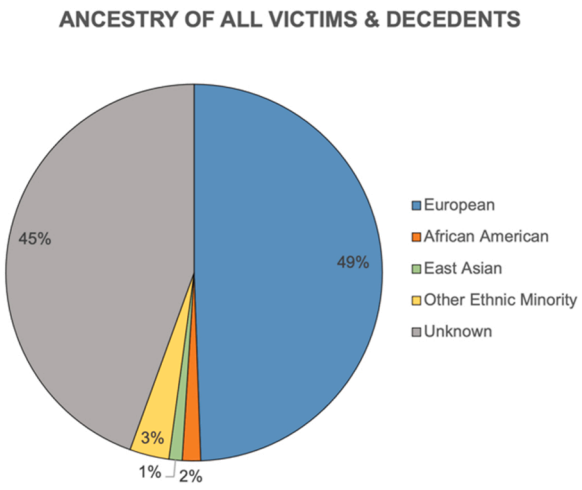


Fig. 17. Ancestry of all victims & decedents. This figure includes data for all cases, from all countries, and for each victim-offender pair, out of a total of 443 victims and decedents.

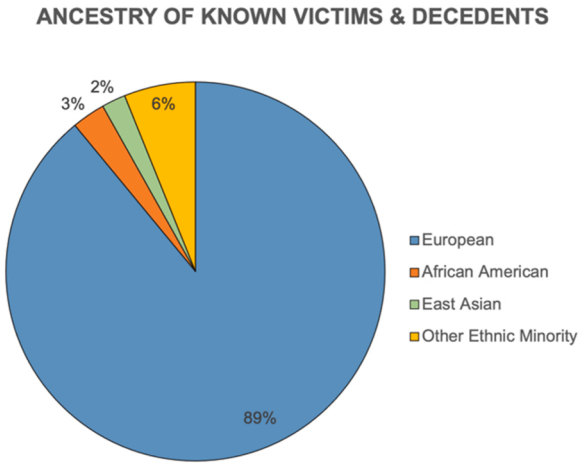


Fig. 18. Ancestry of known victims & decedents. This figure includes data from all countries for all victims and identified decedents, out of a total of 246 victims and decedents, where the ancestry has been reported.

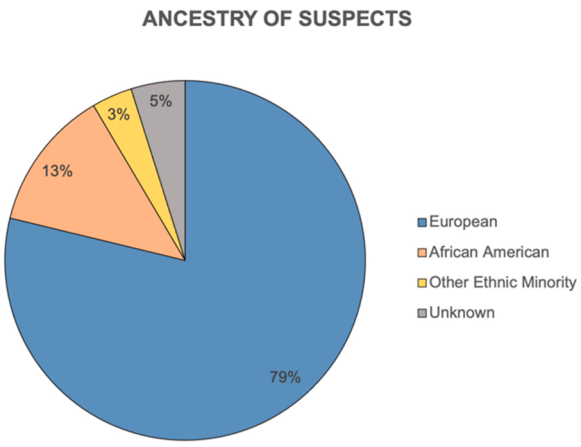


Fig. 19. Ancestry of suspects. This figure includes data for all countries, out of a total of 159 individual suspects. In 6 cases, suspects were identified as having two major ancestries, and these were counted separately for a total of 165 ancestries.

5.6. *Sample source*

Fig. 20 examines data on the biological source of the DNA sample used in the FGG investigation. The most common biological sample used was semen, with about 30% of all samples identified as being from a sexual assault kit, or from a source of semen that was either found at the crime scene, taken from a victim's clothing, or was not specified. The second most common sample used was bone or human remains (with the exact tissue source of the human remains sample left unspecified), for a total of 17% of all cases. In only 2 cases was touch DNA successfully used to obtain a profile of a suspect. In 10 cases, fingernail scrapings were used, but in only half of these cases was it the only sample used.

In a small number of cases, DNA samples from a deceased suspect were available from a law enforcement sample or a medical sample that was already on file. In two cases, the DNA sample was taken from a living person as part of an investigation to identify a living doe (Case IDs 1221 and 1228). There was a large number of cases in which the sample source was unspecified, and so further information needs to be collected before any conclusion can be drawn about the source of the samples used to conduct FGG investigations.

Fig. 21 looks at all U.S. criminal cases that have been cleared using FGG, by individual offender, to determine the method by which the initial results of the FGG investigation were confirmed by police investigators. The data was broken down by whether the suspect was alive

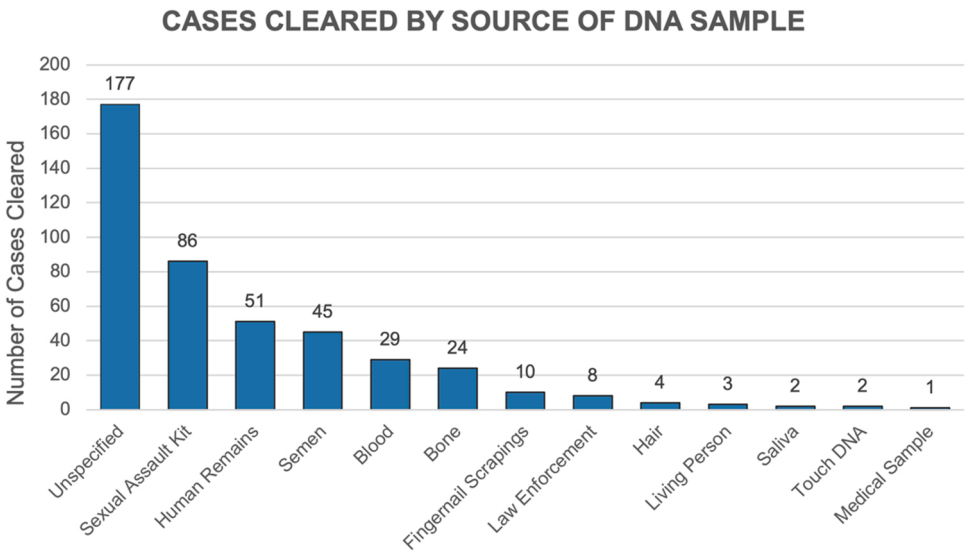


Fig. 20. (a) Cases solved by source of DNA sample. (b) This figure indicates the biological source of the sample used in the FGG investigation, for all cases and all countries, out of a total of 442 cases, including 15 cases in which 2 samples were identified, and 3 cases in which the data was not applicable.

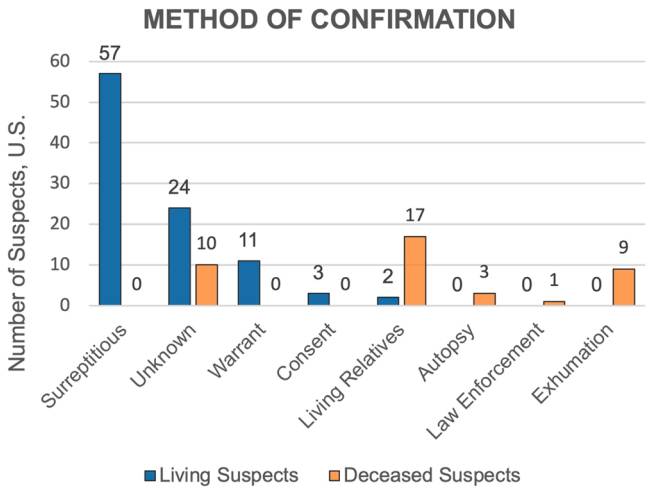


Fig. 21. Method of confirmation. This figure looks at all U.S. criminal cases that have been cleared, by individual offender, to determine the method by which the initial results of the FGG investigation were confirmed by police investigators, out of a total of 137 suspected perpetrators for whom data was available.

at the time that their DNA was sampled and compared to the crime scene DNA, or whether the suspect was deceased.

For living suspects, the most common method of sampling was surreptitious sampling (42%), whereas for deceased suspects and decedents the most common method was a kinship match with a close living relative (43%), followed by an exhumation order (23%). For living suspects, surreptitious sampling (57 cases), a warrant (11 cases), or a sample obtained with the suspect's consent (3 cases) were used to obtain a forensic STR profile suitable for uploading to the CODIS database, and to confirm that the suspect was included as a possible contributor to the crime scene DNA.

5.7. Cases of wrongful prosecution

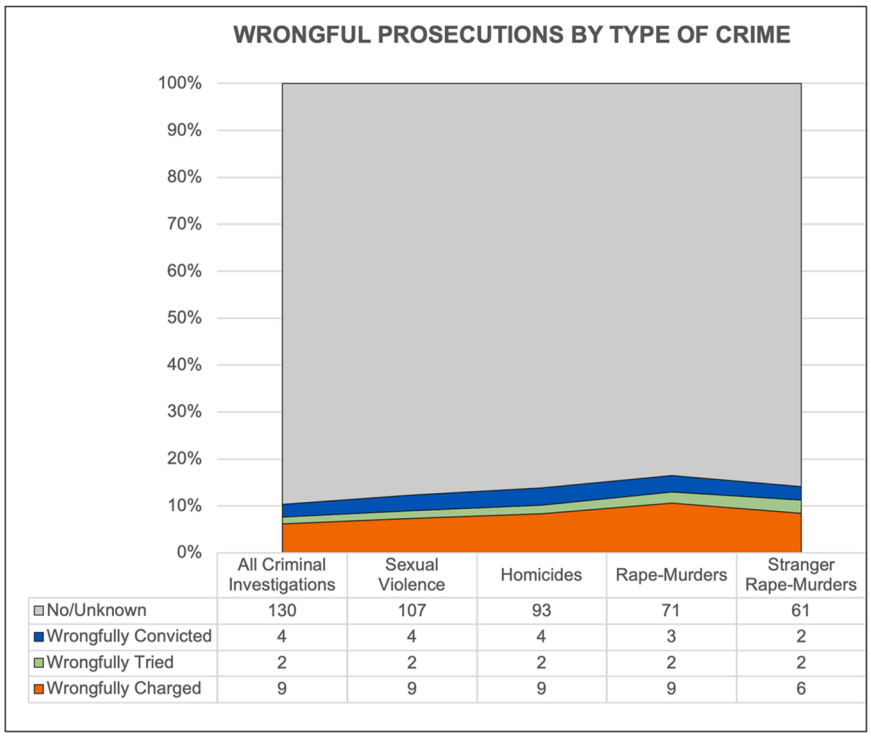
Fig. 22 examines data on wrongful prosecutions against suspects who were later cleared of involvement in the crime. There are a total of 9 individuals who had been wrongfully charged with the crime and were detained and later released, 2 persons who were put on trial for the crime but were later acquitted or had the charges withdrawn, and 4 individuals who were wrongfully convicted of the crime that was later solved through FGG. This data focused on published reports, and so may underestimate the number of innocent suspects who were arrested and charged, especially in older cases.

Rates of wrongful prosecutions in the sample were higher for homicides motivated by a sexual assault (rape-murders). Rates of wrongful conviction in the sample range from 2.8% of all criminal investigations, to 3.5% for all rape-murders, and to 3.7% of all homicides. Rates of all wrongful prosecutions range from 10.3% of all crimes, to 14.1% for all stranger rape-murders, and 16.5% of all rape-murders.

Two of the four individuals who were wrongfully convicted were exonerated through the FGG investigation. One case (Case ID 1334) involved the wrongful conviction of Ricky Davis for the July 7, 1985 homicide of 54-year-old Jane Hylton in El Dorado Hills, California. Davis was convicted in 2005, and he served almost 15 years of a 16-year-to-life sentence for second degree murder [42]. He was convicted despite the fact that his DNA did not match the profile developed from Hylton's remains. His conviction was based on the false testimony of Connie Dahl, who was living in the home with Davis and Hylton at the time of the murder and who stated that she helped Davis commit the murder [42]. Davis was granted a re-trial, and he was being held in custody while the case was pending; he was only released when Michael Green was identified as the lead suspect through FGG [42]. Green, who was 17 years old at the time of the murder, was arrested on February 11, 2020 and charged with homicide [42]. Davis was exonerated with the assistance of the Northern California Innocence Project [42].

The second exoneration (Case ID 1207) involved the wrongful conviction of Chris Tapp who spent 20 years in prison for the 1996 murder of 18-year-old Angie Dodge in Idaho Falls, Idaho. Police had used coercive interview tactics against Tapp that are now known to elicit false confessions [43]. Tapp's DNA was profiled, and he was found to be excluded as a contributor to the crime scene DNA [43]. As with Davis, this did not assist Tapp, as he had falsely told police that, while he did not sexually assault Dodge, he was with two men who did. Tapp was exonerated with the help of the Idaho Innocence Project, and with the support of Angie's mother, Carol Dodge, who was eager to find the truth about her daughter's murder [43].

Brian Leigh Dripps Sr. was arrested on May 15, 2019 and charged with the murder of Angie Dodge. When he was arrested and confronted with the DNA evidence, Dripps made adverse admissions to the police, stating that he did not mean to kill Dodge, just to hold the knife to her



	All Criminal Investigations	Sexual Violence	Homicides	Rape-Murders	Stranger Rape-Murders
No/Unknown	130	107	93	71	61
Wrongfully Convicted	4	4	4	3	2
Wrongfully Tried	2	2	2	2	2
Wrongfully Charged	9	9	9	9	6

Fig. 22. Wrongful prosecutions by type of crime. This figure was calculated by examining the number of innocent persons who were wrongfully charged, tried, or convicted of an offence later cleared through FGG. The metric was calculated per criminal investigation, once per individual offender.

throat; he thought she was still alive when he left [43].

The third wrongful conviction in this sample is that of Guy Paul Morin (Case ID 1416), who was wrongfully convicted of the October 3, 1984 rape and murder of 9-year-old Christine Jessop in Queensville, Ontario [44]. Morin was exonerated in 1995 when a DNA profile of the suspect was created from semen found on Jessop's underpants, making him the first Canadian to be exonerated through DNA testing [44]. The Toronto Police Department and Othram Inc., under the leadership of Anthony Redgrave, identified Calvin Dana Hoover as the lead suspect in Jessop's murder; he was an acquaintance of the Jessop family who lived nearby, and he died by suicide in 2015 [44].

The fourth wrongful conviction in this sample is the unusual case of Henry Lee Lucas, who was convicted in 1984 of the murder of 23-year-old Debra Louise Jackson (Case ID 1233). Jackson's body was found on October 31, 1979, in a drainage ditch off the Interstate-35 highway near Georgetown, Texas [45]. She had been sexually assaulted and strangled and was naked except for a pair of bright orange socks. She was known only as 'Orange Socks' until she was identified in 2019 by the DNA Doe Project [45]. Lucas confessed to killing Jackson – as he confessed to hundreds of other murders – but the evidence supports a finding that many of his confessions were falsely elicited and that police likely fed him crime scene details from their case reports [46].

A 1986 Report by Jim Mattox, then Attorney General for Texas, found that many of Lucas' confessions – including the murder of Jackson – were unsafe because police had used unsound interrogations techniques that were known to elicit false confessions [46]. The Attorney General's investigation also found numerous witnesses who testified that Lucas had been living and working in Florida at the time of Jackson's murder [46].

Lucas' case is also notable as he was sentenced to death for Jackson's murder, but his death sentence was commuted by then-Texas Governor George Bush [47]. Lucas is the only offender on Death Row in Texas whose death sentence was commuted by Bush [47]. Although the murder conviction was never vacated, it appears to be unsound, and the police continue to investigate Jackson's murder. Lucas passed away in prison in 2001 at the age of 64 [47].

Two individuals were tried and acquitted of crimes that have now been attributed to a suspect identified through FGG. Joe Alsip was prosecuted for the March 13, 1980, murders of Lyman and Charlene Smith (Case IDs 1019 and 1028) in Ventura, California [48]. Alsip was bound over for trial after a preliminary inquiry, but the charges were later dropped by prosecutors; he was finally released for lack of evidence in May of 1982 [48]. Joseph James De Angelo was identified through FGG and pled guilty to these murders in 2020 [1].

Richard Bingham falsely confessed to the May 4, 1996, murder of 17-year-old Jessica Baggin in Sitka, Alaska (Case ID 1401). He was tried for the crime and acquitted in 1997 [49]. He is reported to have cognitive disabilities and was coerced into a confession by police, although none of the physical evidence linked him to the crime [49]. Bingham spent 13 months in prison before being acquitted by a jury, after only two hours of deliberations. After his acquittal, Bingham never went back to Sitka. He has stated that it is only now that Steve Branch has been identified as the lead suspect for this crime that the "fog of suspicion that has clouded his life is lifting" [49]. Branch died of suicide when Alaska State Troopers sought to question him for this crime [49].

5.8. *Cases cleared due to identification of decedent*

Fig. 23 examines human remains investigations in which the decedent is a suspected homicide victim. Out of a total of 64 such cases in the sample, 28 (44%) were cleared after the decedent was identified. In most cases, this was due to the close relationship between the victim and the suspect: in only 5 of these cleared cases was the suspect known to be a stranger; in 19 cases, the victim was the child of the suspect, in 4 cases the victim was a parent or a partner, and in 2 cases the victim was an acquaintance. For the cleared cases, 14 are currently pending before the courts, 9 were cleared by exception due to the death of the suspect, and five have resulted in a guilty plea.

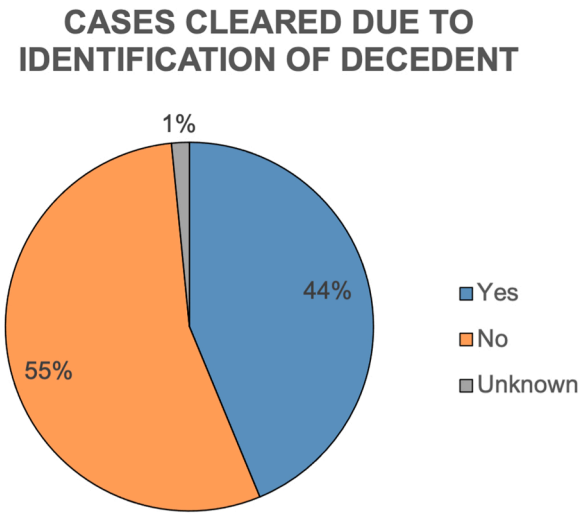


Fig. 23. Cases cleared due to identification of decedent. This figure includes all human remains investigations where homicide is suspected. Out of a total of 64 individual decedents thus identified, 30 homicides had been cleared as of 2020 as a result of the decedent's identification.

5.9. Length of investigations, pre- and post-FGG

Fig. 24 compares the total time for each case to be cleared with the time it took for the FGG investigation to clear the case. Data on when the FGG investigation commenced was only publicly disclosed in 51% of cases. The average time for case clearance in the sample was 23.6 years. The average time for FGG investigations to clear cases was 12.1 months. All FGG investigations cleared to date were completed in less than 3 years, 83% were completed in less than 1.5 years, and 44% were completed in less than one year.

This data looks only at FGG cases that have been successfully cleared. There may be other cases that are actively being researched that have yet to be solved. Without complete transparency on the number of FGG cases that are actively being investigated, it is difficult to gain a complete picture of the probability that a case will be solved, or how long its resolution will take.

5.10. Geographical distribution of cases across population centers in the U.S

Fig. 25 compares the distribution of FGG investigations conducted in the U.S. across 955 Metropolitan and Micropolitan Statistical Areas [24]. The average size of the metro area conducting an FGG investigation was 233, which corresponds to a population of about 180,000 persons. The standard deviation was found to be 350, indicating that there was a wide range of demographic areas using FGG. California was the leading state, with a total of 129 cases cleared, 72 of which were committed by Joseph DeAngelo (Case IDs 1017–1088).

As can be seen from Fig. 25 and Table 4, the distribution of FGG investigations differed from the distribution of the U.S. population as a whole. There was a slight overrepresentation among the mid-sized

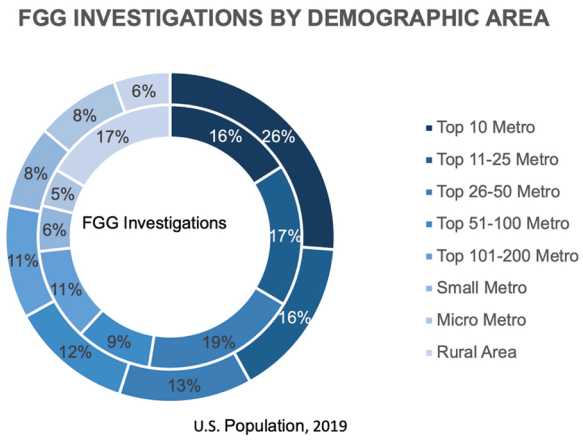


Fig. 25. FGG investigations by demographic area. This figure examines the distribution of FGG investigations across different demographic areas in the U. S., by Metropolitan Statistical Area (MSA). There are 955 MSAs, including all rural areas, which were coded here as '955'. This data is compared with the distribution of the U.S. population across those same statistical areas as of 2019, according to the U.S. Census Bureau. This data includes a total of 430 FGG investigations conducted in the U.S., with criminal cases examined by victim-offender pair.
Source: [24,25].

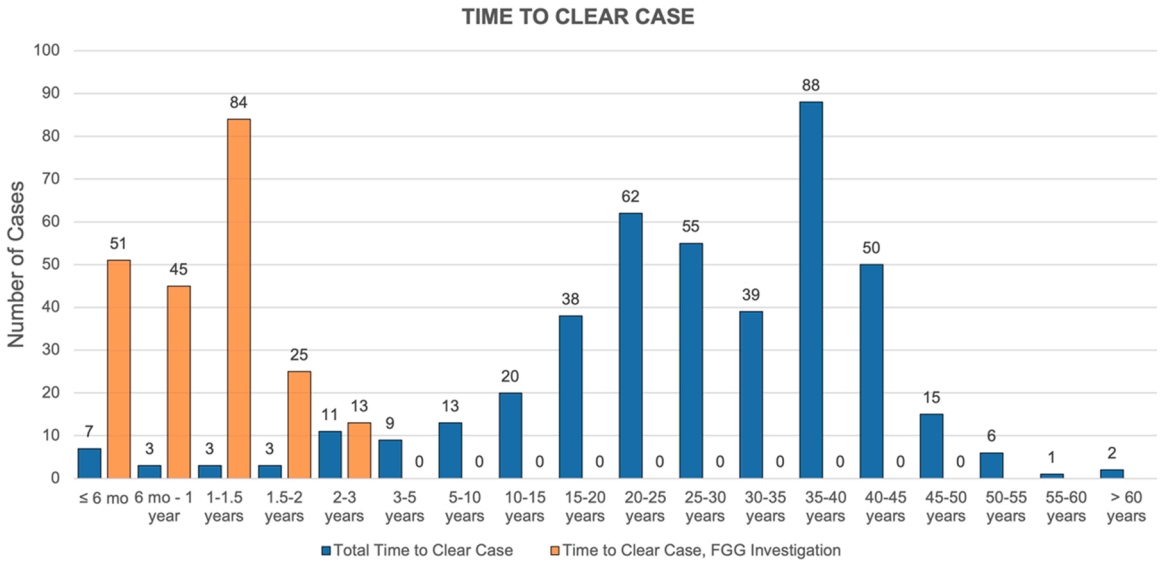


Fig. 24. Time to clear case. This figure compares the total time to clear a case with the length of time for the FGG investigation to clear the case, in years. There are 425 FGG investigations for which the total time to clear the case is indicated, in blue. There are 218 FGG investigations for which data was publicly disclosed as to when the FGG investigation was commenced and concluded, and which are indicated in orange.

metropolitan areas that ranked 26–50 (19% vs. 13% expected), as well as metropolitan areas that ranked 51–100 (9% vs. 12% expected). There was a slight underrepresentation of small metro areas (6% vs. 8% expected) and micropolitan metro areas (5% vs. 8% expected). The largest deviations occurred at either end of the spectrum, with fewer than expected FGG investigations in the 10 largest metropolitan areas (16% vs. 26% expected), and greater than expected FGG investigations in rural areas (17% vs. 6% expected).

Fig. 26 looks at the lead investigating agency that commissioned an FGG investigation, and whether the investigating agency is inside or outside a Top 50 Metropolitan Area, according to the U.S. Census Bureau statistics for 2019 [24]. For investigating agencies that are inside a Top 50 Metropolitan Area, the data looks at whether the investigating agency was the largest police department in that metro area, or a smaller agency in a smaller community within the broader metro area.

The distribution of FGG investigations is very closely aligned with the distribution of the U.S. population that resides inside versus outside of the Top 50 Metropolitan Areas. However, FGG cases were distributed highly unevenly inside the Top 50 Metro Areas. Only 12% of FGG investigations were conducted by the largest police department in a Top 50 MSA. This supports the hypothesis that the largest metropolitan areas are systematically less likely to use FGG, based on their population size, and that within the largest metropolitan areas, the smaller communities and police departments are more likely to use FGG as an investigative tool.

Of the 50 largest metropolitan police departments in the U.S. [50], only 19 conducted an FGG investigation (38%), whereas 31 did not (62%). Of those police departments that used FGG, they cleared a total of 51 criminal cases out of 430 criminal cases in the sample (12%) and arrested and charged a total of 22 individual suspects out of 155 individual suspects in the sample (14%). The discrepancy between the number of cases cleared and the number of individual suspects arrested demonstrated the extent to which those police departments are using FGG to clear cases involving serial offenders.

The ten largest police departments in the U.S. [50] cleared a total of 9 out of 430 cases, and arrested 5 suspects out of 155, indicating that, for this sample, the larger police departments were less likely to use FGG as an investigative tool.

5.11. FGG investigations by demographic area and rates of violent crime

Tables 5 and 6 examine all U.S. cases involving a criminal offence that have been cleared using FGG and compares the rates of crime and number of FGG investigations in the sample with what would be expected given the crime rates as seen in the FBI-Uniform Crime Reports data set concerning violent crimes that took place in 2018 and 2019 [26,27]. This enables the distribution of FGG cases across different demographic areas to be compared with the rates of all violent crimes, homicides, sexual assaults, and robberies in those same areas at about the same time frame as the FGG investigations were being conducted.

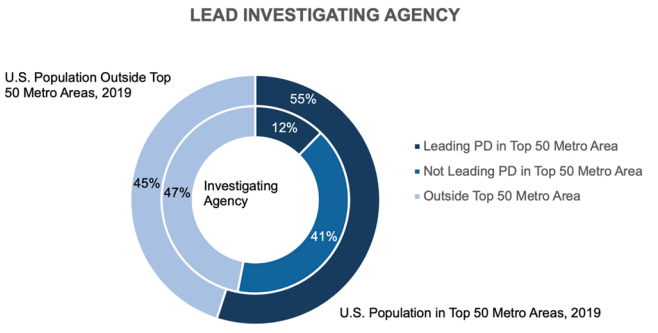


Fig. 26. Lead investigating agency. This figure looks at the lead investigating agency that conducted the FGG investigation, and whether that agency is inside or outside of a Top 50 Metropolitan Area, according to the U.S. Census Bureau statistics for 2019 [24]. For investigating agencies that are inside a Top 50 Metro Area, the data looks at whether the investigating agency was the largest police department in that metro area, or a smaller agency in a smaller community within the broader metro area. This data includes a total of 430 FGG investigations conducted in the U.S., with criminal cases examined by victim-offender pair.

Table 5 looks at how violent crimes, homicides (not including negligent manslaughter), sexual assaults, and robberies were distributed across six demographic areas according to the FBI Uniform Crime Reports data sets for 2018 and 2019 [26,27]. These were averaged to create the probability distributions seen in Table 5. Table 6 compares the expected versus observed number of violent crimes in the present sample based upon these probability distributions and uses a Chi-Squared Goodness of Fit Test to assess the deviation from the expected distribution.

The largest metro counties with populations above 100,000 were using FGG as a strategy to clear violent crimes less often than expected. FGG investigations were underrepresented in all violent crime categories except sexual assaults, which were slightly overrepresented, likely due to the number of serial sexual offenders identified in these jurisdictions. For the smallest population areas – non-metro counties with a population under 10,000 – FGG investigations were overrepresented for all categories of violent crime.

Metro counties with a population of 25,000–99,000 were using FGG more often than expected, and FGG investigations were overrepresented in all violent crime categories. The remaining two categories, metro counties with a population under 25,000 and non-metro counties with a population over 25,000, were using FGG investigations less often than expected for all categories of violent crime.

5.12. Income distribution of counties using FGG

Figs. 27–29 examine data concerning the annual *per capita* personal income of adults in each U.S. county that conducted an FGG investigation, according to the most recent data as found in the 2019 U.S. Census [30]. This data allows for testing of the hypothesis that higher income counties may be more likely to use FGG as they may be prioritizing

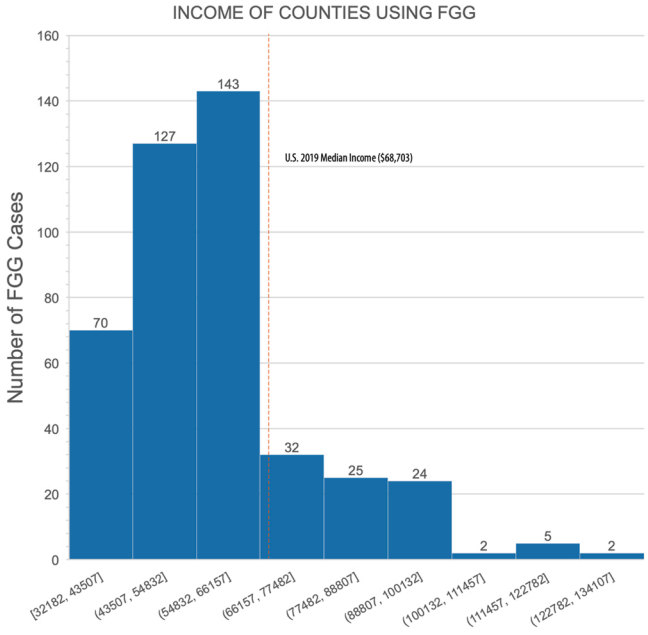


Fig. 27. Income of counties using FGG. This figure looks at all U.S. cases in the sample as compared with the personal *per capita* income of that county, as of the 2019 U.S. Census [30], out of a total of 430 cases. The dashed red line indicates the 2019 U.S. median income of $68,703 USD.

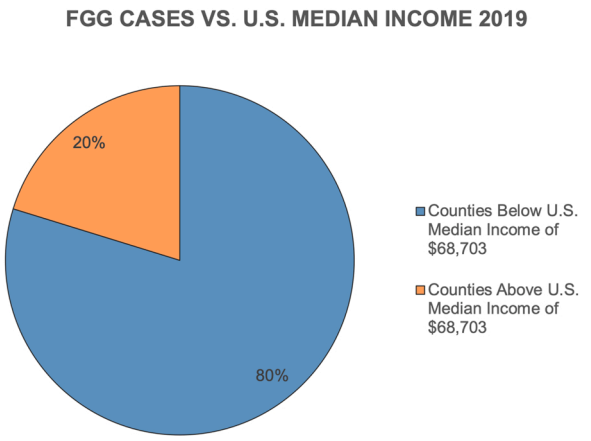


Fig. 28. FGG cases versus U.S. median income 2019. This figure looks at all U.S. cases in the sample and compares the *per capita* income of adults in that county with the U.S. median *per capita* income of $68,703 USD, as of the 2019 U.S. Census, and out of a total of 430 cases.
Source: [30].

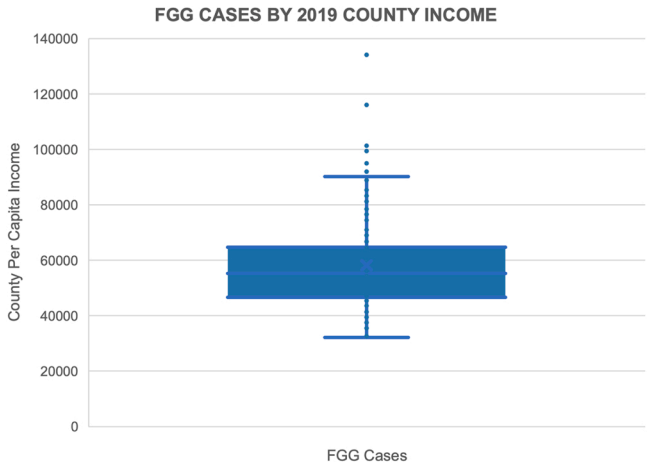


Fig. 29. FGG cases by 2019 county income. This figure shows the income distribution of counties using FGG for all U.S. cases. The average income was $58,135, and the median income was $55,266. Incomes ranged from a minimum of $32,182, to a maximum of $134,107. Only 9 cases out of 430, or 2.1% were conducted in a county with a per capita income over $100,000. Such cases appear on this chart to be outliers.

wealthy victims of crime over poorer victims [51], or because they have greater resources available to them.

As can be seen, 80% of all FGG cases took place in counties where income was below the 2019 U.S. median income of $68,703, and only 20% of cases were conducted in counties with an income above the median. The average income of counties using FGG was $58,135, and the median income was $55,266. County incomes ranged widely, from a minimum of $32,182, to a maximum of $134,107. Only 9 cases out of 430, or 2.1%, were conducted in a county with a per capita income over $100,000. It therefore appears that FGG investigations are concentrated in low- and low-to-moderate-income counties that are below the median income and are not often used in counties with the highest incomes.

6. Discussion

6.1. Substantive justice

Substantive justice is not achieved when violent crimes remain unsolved. Harms against victims go unaddressed, communities feel unsafe, and recidivist offenders pose a danger to public safety. For accused persons, evidence is degraded or lost, witnesses pass away or their memories fade [9], suspicions in the community remain unchecked, and wrongful arrests and even convictions can take place.

Studies demonstrate that unresolved violent crimes take a toll on victims and survivors, their families and friends, and the community as a whole [52]. The community suffers when offenders of serious crimes are not held accountable, and public safety is not ensured; this impacts the trust that people place in the criminal justice system and the rule of law. For victims and survivors, they must deal with grief and loss, pain and suffering, recovery from physical and emotional injuries, a fear of crime, a reluctance to be out of their home, and the fact that neighborhoods deserted of productive public activities leave more room for crime to flourish [52]. Rape can produce very severe trauma, causing multiple, long-term negative outcomes for survivors [53]. Fear, anxiety disorders, drug and alcohol use, and suicidal thoughts are commonly seen in survivors [53]. It has also been shown that negative outcomes are promoted by cultural beliefs and stereotypes that blame women, that promote self-blame, and that prioritize the interests of male perpetrators over those of victims of sexual violence [53]. Social acceptance of sexual violence is therefore a strong predictor of negative outcomes [53].

For these reasons, victims of sexual violence should not be encouraged to subordinate their interests to those of the perpetrator or his or her relatives. One legal scholar has stated, in the context of the related technique of familial searching for relatives in CODIS, that family members of suspects are harmed when these crimes are solved, and that our commitment to solving violent crimes "should be balanced against the claim to freedom from such victimization possessed by innocent relatives in equal proportion to the innocent victims of crime" [22]. Murphy states that it is traumatizing for family members to have a loved-one convicted of a serious crime, such as a serial rape or serial killing. She states that this argument "is not to trivialize the victim's interest in finality and closure, but merely to acknowledge that it can come at the expense of inflicting trauma on other innocents" [22]. Similarly, Moran has stated that the right of victims to receive justice must be balanced against the privacy interests of perpetrators and their relatives [54].

Homicide, like rape, is especially traumatic for families, communities and other survivors coping in its aftermath. McDevitt-Murphy has shown that survivors of homicide faced "significantly more severe PTSD symptoms and general distress" than a control group with no trauma exposure as well as a group that had been exposed to non-homicide trauma [55]. Survivors' trauma and distress was strongly related to their relationship with the victim – with African Americans being more affected than other groups, and mothers of homicide victims being the most distressed and most likely to have symptoms of post-traumatic stress disorder [55]. Homicide is also harder to grieve than other forms of loss: it has been shown to cause more distress and a more difficult recovery than for those who lost loved ones to other causes [55]. The authors conclude that "individuals surviving the homicide of a loved one are at considerable risk for adverse psychological outcomes in multiple domains" [55].

The family members of recidivist and sexually violent offenders may not be best served by allowing those offenders to go unidentified and unhindered in their criminal activities. This dataset contains several suspects who are accused of physically or sexually abusing their partners, children, or other close family members (Case IDs 1091–1092, 1130, 1157–1158, 1212–1221, 1223, 1300–1301, 1369–1372, 1393–1394), and 7 suspects who are accused of killing more than one close family member, such as a partner or a child (Case IDs 1157–1158, 1184–1185, 1212–1221, 1328–1329, 1330–1331, 1393–1394, 1441–1442).

Berkman has argued that we might consider using FGG only to clear cold cases, due to privacy concerns [56]. There may be good reasons to use FGG only when other methods of forensic DNA analysis using law enforcement databases are not useful. On the other hand, there are also

good reasons why violent crimes should be solved before cases go cold. Many of the cases here languished for many decades before being solved. The harms of violent crime are compounded when crimes remain unsolved for long periods of time. The U.S. National Institute of Justice (NIJ) states that there is a crisis in the number of unsolved violent crimes in that country [52]. Clearance rates for homicides in the U.S. have fallen in recent years, while clearance rates for sexual assaults remain quite low; the result is that about 63% of perpetrators of sexual assaults and about 40% of perpetrators of homicides are never identified or held accountable [52]. In the U.S., about 242,000 unsolved homicides have accumulated since 1980 and each year there are about 6000 more [57]. There are also about 40,000 decedents awaiting identification [57]. In 2019, the FBI's Uniform Crime Reports indicated that only 45.5% of all violent crimes were cleared, including 61.4% of all homicides, and 32.9% of sexual assaults [29] A large part of the cold case crisis is due to serial and recidivist offenders [52].

Clearing violent crimes helps to build public trust in the criminal justice system and the rule of law by letting the public know that victims will be heard, laws will be enforced, and perpetrators will be held accountable [52]. The NIJ states that "not producing results jeopardizes any community's goodwill and the positive public perceptions that exist. Resolving cold cases is an important contributor to preventing crime, increasing public safety," and promoting trust [52].

The random nature of serial killings, stranger murders, and serial rapes generates a good deal of fear in the local community [41]. Crimes in which the perpetrator is a stranger to the victim are also among the hardest to solve, as evidence, witnesses, and motives that are present in other cases are absent [58]. Many perpetrators take steps to avoid capture – such as hiding the bodies, cleaning the crime scene, avoiding police contact, failing to attend court hearings, or not providing DNA samples or registering as sex offenders when so ordered by a court. This dataset contains 6 suspects who were convicted sex offenders, but whose DNA samples were not obtained due to official error or the offender's absconding (Case IDs 1106, 1191, 1231–1232, 1312–1313, 1415, 1417). In one case (Case ID 1106), Michael Henslick had twice failed to comply with a judge's order that he provide a DNA sample in the years following his murder of Holly Cassano [59]. This allowed him to remain at large for nine years.

There have been several cases reported elsewhere which show that clearing DNA backlogs and ensuring that DNA orders are enforced can result in more crimes being cleared through traditional forensic DNA methods [59]. However, the American Law Institute (ALI) has recently adopted several amendments to the Model Penal Code's provisions on sexual assault registration [60] that, if adopted, may make it more difficult to identity sexual offenders in the United States and hold them accountable when they fail to register [61,62]. These changes include a recommendation that charges for failing to register be prosecuted as a misdemeanor (s 213.11G), as well as removing the requirement that convicted sex offenders provide identification in the form of fingerprints and DNA samples (s. 213.11D) [60].

Those who are targeted for sexual violence by serial offenders are chosen based upon their "availability, vulnerability, and desirability" [63]. Serial rapists tend to prey upon strangers, and select their victims in a strategic way, looking for those who are vulnerable and for opportunities to avoid apprehension [10,64]. These offenders are often ambitious, intelligent, and driven to dominate others [40]. Crimes of interpersonal violence require a high degree of physical fitness and mental acuity, and so are most often perpetrated by younger men [65]. Rates of sexual offending tend to decrease with age for this reason [66].

The cases that have been solved to date using FGG fit this picture: 28% of the victims belonged to groups that are especially vulnerable to criminal violence and sexual exploitation, including children and teenagers, the elderly, sex workers, those with cognitive or developmental disabilities, and gender non-conforming persons. In contrast, few of the suspected perpetrators belonged to these groups: only 8% were classed as vulnerable because they were teenagers at the time of the offence, and none belonged to any other vulnerable group.

The suspects were overwhelmingly young males under the age of 35, primarily of a European ancestry, and physically and mentally fit at the time of the criminal offence. About 11% of suspects belonged to public security forces such as the police or the military; because many were serial offenders, they constituted 29% of all criminal investigations in the sample. In a small number of cases this may have played a role in the suspects' offending or shielded them from scrutiny. Less than 1% of victims were so identified, and in one case a female's training as a police officer enabled her to successfully fight-off a serial home-invasion rapist (Case ID 1107), who then proceeded to target elderly women (Case IDs 1108–1110).

In contrast, the victims of crimes in this sample were overwhelmingly female (84%), were either very young or very old or belonged to another social group that is especially vulnerable to criminal victimization and sexual exploitation (28%). Many victims in the dataset were attacked in their homes, or while walking alone. Victims also tended to be from U.S. counties with personal incomes below the U.S. median income (80%), with very few victims from counties with high incomes (over $100,00 USD) and thus likely to be of higher socioeconomic status (2.1%). Of the male victims, many were victimized as part of a sexual assault against a female partner or relative, or were children killed by a parent; others were victims of robberies while at their home or workplace. A small number were victimized by a pedophile who targeted young men and boys. Therefore, many male victims in the sample were in a vulnerable position when they were attacked, as well.

Since homicide and sexual assault are crimes with high rates of recidivism, public safety is best served by identifying and apprehending the perpetrators of these crimes. A review of the literature concerning the rates of recidivism of homicide offenders after a period of incarceration found that those who committed a second homicide upon release from prison were rare, with rates of specific recidivism ranging from 1% to 3.5% [67]. One study of sexual homicide offenders found that about 3.3% out of 139 such offenders were reconvicted of an actual or attempted homicide [68]. There are many reasons for these low rates of recidivism. Homicide is a relatively uncommon crime, homicide perpetrators tend to offend less often with advancing age, due to the physically strenuous nature of the offending, particularly for sexually motivated homicides [68]. Offenders may be less likely to reoffend when they have been previously convicted and are being monitored as part of a parole or reintegration program. Other studies have found that homicide recidivism is increased in those with a personality disorder, as well as those who use drugs or alcohol [69].

Serial rape is not a common crime, but because serial rapists have multiple victims these offences account for a significant proportion of all sexual assaults reported to police [10]. Serial rapists are also relatively more difficult for police to identify [10]. Serial rapists are usually strangers to the victim, they tend to be quite sophisticated, and they become better able to avoid apprehension with experience [64]. It can be difficult to measure rates of recidivism for sexual offenders, since methodologies vary, and many assaults are not detected and do not appear in official data [66]. The 15-year recidivism rate for sexual offenders who were re-arrested for a sexual offence has been found to be around 37% [66]. One study followed sex offenders who were released from prison over a 9-year period and found that about 7.7% were rearrested for another rape or sexual assault [70]. Recidivist sex offenders accounted for about 16% of all arrests for sex offences in that same time frame [70]. Sex offenders who choose children as their victims have some of the highest rates of recidivism, especially those who victimize boys [66].

The rate of recidivism for sexual offenders and serial murderers in the dataset was relatively high. Rates of recidivism were 31% for homicide offenders, which is significantly higher than the rates of 1–3.5% seen by Liem and Hill [67,68]. The rate of recidivism for sexual offenders in the data set was 37%; this, too, is higher than the rates of

recidivism seen by Przybylski and Alper [66,70], and closer to the highest rates seen in sexual offenders after a 15-year period. This supports the hypothesis that law enforcement agencies have been using FGG to target recidivist murderers and serial sexual offenders. It also supports the hypothesis that when offenders remain unhindered in their criminal career for long periods of time then the probability of their committing similar crimes is correspondingly higher.

Serial offenders who remain wrongfully at large cause significant damage to the community, both because they kill further victims, and because innocents are more likely to be wrongfully convicted of their crimes [71]. Many of the serial offences in this dataset would have been prevented had the initial offence been cleared expeditiously. There were 28 known or suspected serial murderers in the dataset who are thought to be responsible for 95 murders, as well as sexual assaults or violent attacks on 76 further victims. Had these offenders been apprehended after their first known crime, about 67 lives may have been saved. There were also 64 known or suspected serial sexual offenders in the sample who are responsible for sexual assaults against 308 known victims. Had these offenders been apprehended after their first crime, about 244 of these sexual assaults could have been prevented. These harms should be factored into a proportionality calculus in which the harms of an investigative technique are weighed against the harms of leaving perpetrators wrongfully at large.

6.2. Procedural justice

When violent crimes remain unsolved for long periods of time this has a negative impact on suspects and accused persons. Evidence – both inculpatory *and* exculpatory – is degraded or lost, witnesses pass away or their memories fade, suspicions in the community remain unchecked which can blight the lives of innocent persons, police request DNA samples from members of the public which places them under suspicion and violates their genetic privacy, and individuals can even be wrongfully arrested, prosecuted, or convicted.

The rights of defendants to a fair trial are best secured when cases are cleared quickly and brought to trial in a timely fashion. Many of the cases in this sample were very old: 88% had been open for more than a decade, and 38% were open for longer than 35 years. The FGG investigations that have been cleared took an average time of about 12 months. All FGG investigations were completed in less than 3 years, 83% were completed in less than 1½ years, and 44% were completed in less than one year. Greytak has stated that about 80% of the more than 250 cases referred to Parabon Nanolabs have a third cousin match or closer in the D2C database, and thus a significant chance of being cleared [6]. Greytak also found that about 80% of cases involving a DNA sample from an individual of European ancestry were assessed as being workable, compared with about 60% for those of a non-European ancestry, due to the under-representation of those of non-European ancestries in the D2C databases [6].

When the actual perpetrator of a crime is not identified then innocent persons are at risk of becoming suspects. As Murphy states, “the worst indignity of an investigation can be living under a cloud of suspicion; even mere suspicion, quickly dispelled, has the potential to disrupt a career, destroy a marriage, or ruin a life” [22]. As Zabel states, the “clouds of suspicion these innocent individuals live in are dark and noxious,” and suspicion in cases of rape and murder can be very damaging [2].

Over the decades in which the crimes in this sample remained unsolved, many innocent persons came under a cloud of suspicion and had their lives blighted – some severely. A wrongful conviction is defined as a factually innocent person who is convicted of a criminal offence committed by another perpetrator [72]. Wrongful convictions are very harmful, and can cause significant stigmatization, trauma, and financial hardship, even apart from the great wrong of a period of imprisonment. For the crimes in this sample, four men were wrongfully convicted; one, Henry Lee Lucas, was sentenced to death, but this sentence was later commuted [47]. Wrongful convictions also cause trauma and hardship for suspects, victims, and their family members; when a conviction is overturned, victims and survivors come face-to-face with a lack of closure and the knowledge that the true perpetrator is still at large [72].

There have been numerous attempts in the literature to assess the rate of wrongful convictions. Since the actual guilt or innocence of any individual is precisely what we do not know and precisely what the criminal process is intended to adjudicate [73], most scholars have used the number of exonerations as a proxy for the number of wrongful convictions [74]. Exonerations have decreased since their high in the 1980 s and early 1990 s, reaching their height in 1995 [75].

Cases involving rape and murder may have higher rates of wrongful conviction than violent felonies as a whole, partly because there is tremendous pressure on police and prosecutors to secure a conviction [72]. Risinger looked at capital rape-murder cases in the U.S. in the 1980 s, and he found a wrongful conviction rate for the factually innocent of about 3.3%, to a ceiling of about 5% [76]. These cases are similar to the sexually motivated homicides in the sample, which took place in a roughly similar time period in which such crimes were more common than today. The wrongful conviction rate found by Risinger is very similar to the rate of wrongful convictions for rape-murders seen in the present dataset, which is 3.5%.

In the period from 2018, 2019, and 2020, the U.S. National Registry of Exonerations (NRE) recorded 72, 80, and 59 exonerations for homicide respectively, for a total of 211 exonerations [77]. DNA exonerations in the same period were 23, 20, and 17, for a total of 60 [78]. This dataset saw two homicide exonerations in that same time period. This is a small number but nevertheless constitutes almost 1% of all exonerations, and 3.3% of all DNA exonerations. FGG, although a new technique, can make a significant contribution to exonerating the innocent through DNA technology – especially those who, like the exonerees here, are convicted despite the fact that they have been excluded as contributors to a crime scene DNA sample.

Several individuals were wrongfully arrested, charged (Case IDs 1254, 1256, 1257, 1258, 1259, 1305, 1326, 1412, 1429), and even put on trial (Case IDs 1019, 1028, 1401) for murders committed by a suspected later identified through FGG. Several persons were long suspected of crimes committed by others, many of whom were partners or close relatives of the victim and who were only cleared of suspicion when the case was finally cleared (Case IDs 1186, 1193, 1263–1326, 1307, 1374, 1411–1412). The rate of all wrongful prosecutions in the dataset ranges from 10.3% of all crimes, to 16.5% of all rape-murders.

Cases cleared through FGG investigations are producing evidence that, to date, has been admitted by courts and is assisting them in their decision-making. Accused persons have thus far made adverse admissions concerning the accuracy of the DNA evidence in about half of all criminal cases, and this figure is about 1/3 for all individual accused persons in the U.S. (with the difference resulting mainly from multiple victims of serial offenders). Of the live cases that are currently before the courts, about 64% are pending, 22% have resulted in a guilty plea, and 14% have resulted in a conviction at trial. This is higher than the rates of conviction for all felony murders in the U.S., which the Bureau of Justice Statistics has reported as being about 70% [79], and for stranger sexual assaults, which Lundrigan has found is about 75% [80]. Only one accused has had the charges against them dismissed (Case ID 1261). This was not due to FGG having ascertained a factually innocent suspect, but in return for the individual’s cooperation with the prosecution.

The DNA samples used in the FGG cases in this dataset were generally of high quality, being mainly from blood, semen, or human remains, and not from unresolved mixtures. Semen taken from a crime scene or a sexual assault kit was by far the most common sample used in an FGG investigation. Greytak has stated that at Parabon Nanolabs, one of the largest forensic organizations to perform FGG, about 79.4% of all samples were single source, about 16.4% were low mixtures, and 4.2% were from deconvoluted two-person mixtures with a major and minor contributor [6]. To the extent that samples used in FGG criminal

investigations are from single sources or resolved two-person mixtures, are of high quality and are probative of guilt, then the chances of wrongful conviction should be no greater than for any case involving forensic DNA STR profiles and will likely be lower. Further research into the type and quality of DNA samples used in FGG investigations should be performed to collect more information on this important issue.

The DNA evidence that is prepared and used as evidence in court in FGG cases is the same forensic STR profiles that have long been used, and not the SNP profiles used for searching in genealogical databases, as some have wrongly assumed [81]. This kind of DNA evidence, while it can be misused and misinterpreted [82], has proven to be a highly reliable form of evidence, especially when compared with other kinds of evidence that courts and juries use in their decision-making [83]. DNA evidence – when free from contamination, laboratory, or other forms of human error – is reliable and persuasive according to studies involving mock jurors [84]. As the National Institute of Standards and Technology (NIST) has stated in a recent report on DNA mixture interpretation, "When laboratories analyze high-quality, single-source samples, decision-makers often have confidence in DNA test results in part because it has been demonstrated that different laboratories will arrive at the same result" [85]. In the U.S., law enforcement agencies seeking to use FGG must first prepare traditional STR profiles that are uploaded for searching in CODIS, and therefore must meet the quality control guidelines set by SWGDAM [86]. This practice is desirable and should be formalized in regulations and ethical guidelines governing the use of FGG.

Several types of evidence that courts have long relied upon have been shown to be less reliable and to lead to wrongful convictions. The Innocence Project estimates that, based on the first 325 DNA exonerations, about 72% of wrongful convictions involved eyewitness misidentifications, 47% involved invalidated or improper forensic work, 27% involved police eliciting false confessions, and 15% involved unreliable informant testimony [83]. The criminal justice system should be encouraged to use evidence that is reliable and has been scientifically validated. Otherwise, law enforcement may be left to pursue false suspects and leads and will have to rely on less reliable forms of testimony, such as eyewitness identifications and confessions [87].

The risks and benefits of any new investigative technique must be accurately determined and weighed in a proportionality calculus. To date, there have been several misconceptions about how FGG works [81], and very little empirical evidence. Further research into the risks and benefits of FGG would be very helpful in assessing the ethics of its future use and governance. Many scholars have also pointed out the continuing need to clear sexual assault kit backlogs and upload the DNA profiles into national DNA databases [88], and to reduce the backlogs in the collection of DNA profiles from convicted offenders [89].

At present, the Department of Justice Interim Police on Forensic Genetic Genealogical Analysis and Searching requires that forensic DNA profiles first be prepared and submitted to law enforcement databases such as CODIS [11], as do the current SWGDAM recommendations [14]. The Sacramento District Attorney's Office currently requires that agencies first exhaust "all other reasonable and viable investigative leads, including DNA profile comparisons to suspect reference samples, with negative results," and recommends a prior familial search of law enforcement databases [90]. Jurisdictions which currently permit familial searching may decide to recommend or require that this technique be attempted first.

6.3. *Distributive justice*

Achieving distributive justice means that the harms and benefits of criminal investigations should be fairly distributed. Issues have been raised with FGG concerning the privacy of non-suspects [2,5,81], the utility of clearing violent crimes [22], and whether victims from marginalized communities are being well-served [51]. At the same time, there have been several misconceptions about the nature of FGG and how it impacts different groups [81,87].

FGG, like other investigative techniques, involves several trade-offs among the interests of different individuals and groups in society. It is best, therefore, when intrusive techniques are limited to cases that involve a significant threat to public safety. It is also best when the burdens of policing and investigative techniques are borne by the communities in which the crimes being investigated have taken place. This ensures that the communities who bear the burdens of the police investigation are also the ones who will benefit when the offenders are identified and apprehended. Serial killers and serial sexual offenders work to avoid being apprehended and will often look for victims and body disposal sites in areas that they know well, such as communities in which they live or work [10]. The communities in which these crimes take place are often the same communities in which violent offenders live and operate.

Violent crimes tend to be intra-racial, and this is especially true for sex offenders, who preferentially choose victims from their own racial or ethnic group [64]. De Heer's study found that 70% of sexual assault perpetrators were of the same ethnicity as their victims [64]. That is reflected in the present data. Out of 184 cases in which the ethnicity of both suspects and victims are known, 140 (76%) had a shared ethnicity. Of these cases, 138 involved a shared European ancestry. Of those victims who had a non-European ancestry, 15 (71%) were victimized by an offender of European ancestry, 2 out of 21 (9.5%) were victimized by an offender of the same ethnicity, and the remaining 4 (19%) were victimized by an offender of an ethnic minority ancestry that is different from their own.

This research also looked at the geographical and demographic distribution of FGG investigations within the United States. The data collected here demonstrates that rural areas are using FGG more often than expected, whereas the more densely populated metropolitan areas and the largest metropolitan police departments are using FGG less often than expected, given both their share of the U.S. population as well as their share of violent crimes, with the exception of serial sexual offenders.

Since FGG investigations have been overwhelmingly directed against serial and sexually motivated offenders, it might be the case that FGG investigations are roughly following the geographical pattern of these types of offender. Studies of serial killers tend to show that they are distributed roughly along demographic lines, according to the distribution of the American population [41]. However, many serial killers migrate from their place of birth, often in the Midwest, and kill along migration routes, most often in California, Oregon or Washington state, and to a lesser extent in Texas and Florida [40]. Serial killers overall, and sexual serial killers in particular, are less well-represented in the Southern states, especially Georgia, Louisiana, and Alabama [40]. Serial killers also have a preference to dispose of their bodies in rural areas [41]. Organized serial killers in particular tend to plan the disposal of the bodies, and they are more likely to leave their bodies in rural areas [41]. This generally fits the pattern seen in the sample here, but more data needs to be collected to compare the geographical distribution of the offenders in this dataset with similar serial offenders in the U.S. as a whole.

This may partly explain why rural areas were overrepresented in the sample but does not explain why the larger metropolitan areas and police departments are using FGG less often than expected. It may be the case that there are funding constraints, that the technique is novel and controversial, or that higher courts have yet to rule on the admissibility of the evidence. This may also be a result of how FGG investigations are funded. Many of the investigations to date have been funded through public donations, or grants from the Department of Justice Sexual Assault Kit Initiative [91]. It may be that there are barriers to funding in the larger metropolitan police departments, as well as a larger backlog of unsolved cases. It may be that smaller departments can distribute their resources among a smaller number of cold cases than large metro police departments. This is an area that requires further research.

This research also looked at the ancestries of perpetrators and victims. The evidence discloses that victims of European ancestry are better represented in the dataset, and that victims who are of African American, Central and South American ancestry (corresponding roughly to the U.S. Census category of non-white Hispanics), as well as Indigenous persons are less well-represented. This is certainly true for the homicide cases in which the ancestry of the victims is generally known. As of the end of 2020, there was one case that had successfully identified a gender non-conforming individual. As this research is not able to capture pending FGG cases, it is not known how many cases involving ethnic minority victims are currently being investigated through FGG but remain uncleared.

Horowitz et al. have shown that those who are interested in taking a consumer DNA test to discover their ancestry tend to be those who lack important information about their ancestry [92]. Those who have immigrated within the last two generations are less likely to take an ancestry test, as are those of homogeneous ancestry, and Latino/a ancestry [92]. Those who are more interested in taking a DNA ancestry test tend to be those whose ancestors immigrated more than two generations ago, are of mixed ancestry, are older and have higher levels of formal education [92]. Those whose DNA profiles are in the D2C databases used for FGG are likely to show similar characteristics, and this will determine which persons of interest are more readily identifiable.

When making decisions to use FGG, public authorities should be mindful of the needs of victims who have not been well-served by the criminal justice system, including African American and Indigenous women, non-white Hispanic women, persons engaged in sex work, trans and gender non-conforming persons, as well as men and boys who are victimized by serial sexual offenders. These communities should be consulted about the use of FGG investigations to clear crimes that are impacting their community.

This research also looked at the *per capita* income of counties in which an FGG investigation was conducted. It has been proposed that FGG might be a technique used by wealthier counties to solve crimes involving victims of higher socio-economic status [51]. Since FGG requires funding for DNA laboratory work as well as the costs of the genealogical analysis, it may be that wealthier counties are better able to access funding and resources. On the other hand, younger women under the age of 35 years and who are living in more rural and lower-income households are more likely to be victims of sexual violence. Based on this, we might expect more FGG investigations in poorer counties [64].

The evidence in the dataset supports the latter hypothesis. About 80% of all FGG investigations took place in counties that were below the 2019 U.S. median per capita income, and few cases (2.1%) took place in counties with a per capita income of over $100,000 USD. FGG investigations are primarily being used in criminal investigations involving victims living in counties of lower and lower-middle socio-economic status.

7. Conclusion

FGG has been used to date to clear cases involving sexual offences, sexual violence, and to identify recidivist and serial offenders. For most investigations, the sample source used was semen or blood. Therefore, FGG has primarily been used in crimes of intimate personal violence, when there was a very close physical connection between the perpetrator and the victim; crimes primarily took the form of sexual assaults, rape-murders, violent robberies or home invasions, and infanticides.

Victims tended to be persons from especially vulnerable social groups, and were primarily female, very young or very old, and from lower-income counties. The victims tended to be of European ancestry; victims who were African American, non-White Hispanic, and Indigenous persons have been less well-represented in FGG investigations to date. This could be because larger metropolitan areas are using FGG less often than expected – given both their share of the U.S. population, as well as their share of violent crime – whereas rural counties were using FGG more often than expected. Suspects, on the other hand, tended to be of European ancestry, and to have been young as well as physically and mentally fit at the time of the crime. Suspects in this dataset were about 10 times more likely to have killed more than one person than other convicts of felony murder.

Most cases remain pending before the courts. Of those cases that have completed as of the end of 2020, most have resulted in a guilty plea, or a conviction at trial, and none have been acquitted. No court has yet ruled to exclude the DNA evidence obtained, or the evidence of identity as disclosed by the FGG investigation. As most cases before the courts have yet to complete, and appellate processes can be lengthy, this is data that should be monitored over time to determine whether evidence obtained through FGG is assisting courts in their decision-making and leading to the identification of factually guilty persons and is not resulting in wrongful convictions.

FGG evidence to date has assisted in exonerating two individuals who were wrongfully convicted. Many of the cases in this dataset were open for very long periods of time. Rates of wrongful conviction in the sample were comparable to what have been seen in similar cases: rates ranged from 2.1% of all crimes, to 3.8% of all rape-murders. Rates of all wrongful prosecutions of all types ranged from 10.1% of all crimes, to 16.5% of all rape-murders.

FGG cases have preferentially been used to identify serial and recidivist murderers and sexual offenders, particularly in cases where the victim was a stranger to the suspect. These cases involve quite sophisticated offenders and – along with the lack of connection with the victim – are generally harder to clear. FGG cases are being cleared much more expeditiously than were the traditional investigations. The average time for FGG investigations to clear cases was 12.1 months. All FGG investigations completed in less than 3 years, 83% were completed in less than 1.5 years, and 44% were completed in less than one year. FGG therefore has the potential to benefit accused persons in the exercise of their rights to a speedy trial, by ensuring that cases are cleared and brought to court expeditiously. Clearing cases quickly also leads to lower rates of repeat offending, limit the number of persons who are suspected of the crime, as well as those who have their DNA sampled by investigators, or who are wrongfully arrested, tried, or convicted of the crimes.

The public as a whole has a strong interest in reducing violent crime, in assisting victims, in avoiding wrongful convictions, in bringing reliable and probative evidence before the courts, and in protecting privacy rights during criminal investigations. These interests are broadly shared across diverse groups and members of society. In some ways, too, the interests of victims, offenders, and the public at large are incommensurable: the interests of violent offenders in avoiding capture conflicts with the rights of victims to seek justice, as well as the right of future victims to their security and bodily integrity; the failure to clear a case in a timely and accurate manner can expose multiple others to suspicion, to police investigations, and to demands for DNA samples; shielding one family from investigation leaves another family in grief and limbo; for every true perpetrator who is not identified, others are exposed to the risks of violent offending, and even to wrongful arrests and convictions. As Wickenheiser states, the ethical question to be explored is “to what extent can the rights of the innocent general public and relatives of the committer of a crime be infringed upon by examining their genetic data to identify the crime perpetrator and thereby prevent future crimes and improve public safety?" [93]. Different communities may prioritize different interests depending on their community’s current needs and values.

But there is also much common ground. Criminologists now recognize that the long-standing “crime control” versus “due process” model sets up a false dichotomy that does not necessarily serve the needs of persons who come into contact with the criminal justice system [94]. Members of the public have rights to life, to bodily integrity, to seek justice for crimes committed against them, and to receive information about deceased family members. Perpetrators have a right to a fair and

speedy trial, and these rights are not advanced when cases remain in limbo for years, or even decades: witnesses pass away, memories fade, and evidence – both inculpatory *and* exculpatory – gets lost, degraded, or contaminated. Leaving cases unsolved for long periods of time does not advance the goals of procedural justice, nor does it promote decision-making by the courts through the preservation and presentation of high-quality evidence.

FGG can therefore play an important role in promoting substantive, procedural, and distributive justice. As the use of FGG becomes more common, it will and should be further regulated to assure the public that their personal information is treated with respect, and that accredited laboratories and genealogists are carrying out the work. The U.S. states of Maryland and Montana have recently passed bills that require judicial approval of FGG investigations [95,96], and some steps are being taken to draft guidelines for best practices for FGG [10,12,92]. New Haven University's Henry C. Lee College of Criminal Justice and Forensic Sciences has recently launched a graduate certificate program in Forensic Genetic Genealogy to train future practitioners of FGG [97]. Additional steps can be taken to ensure that there is adequate funding for FGG investigations, particularly in the largest metropolitan areas, and that victims from marginalized communities are being well-served. The Trans Doe Task Force is advocating for the interests of trans and gender non-conforming persons, and the special interests they have in their identification [98]. Those public bodies who are making decisions to conduct or to fund FGG cases should be mindful of victims whose interests have been less well-served by the criminal justice system, such as victims of non-European ancestry, trans and gender non-conforming persons, and male victims of sexual violence.

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